# Learning a Codebase

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# Contents


**1 Repository** **3**
- 1.1 Glossary - Summary . . . 3
- 1.2 Repository Summary Table . . . 4
- 1.3 Sample scatter touched against commit - . . . 5
- 1.4 Total commits by developer - Developer Commits . . . 6
- 1.5 Histogram components touched by developer - For components with total . . . 8
- 1.6 Time series components touched - For all components touched for all commits and for each period . . . 17
  - 1.6.1 Time series components touched - For packages touched for first five commits . . . 17
  - 1.6.2 Time series components touched - For packages touched for first ten commits . . . 22
  - 1.6.3 Time series components touched - For packages touched for first twenty commits . . . 27
  - 1.6.4 Time series components touched - For packages touched for commit . . . 30
  - 1.6.5 Time series components touched - For packages touched for each week commit . . . 35
- 1.7 Welch t test - For all components touched for all commits . . . 40
  - 1.7.1 Welch t test - For all components touched for all commits by Founder(74) against Joiner 41
  - 1.7.2 Welch t test - For all components touched for all commits by Moderate(620) against Sustained . . . 42
  - 1.7.3 Welch t test - For all components touched for all commits by Moderate founder(40) against Moderate later joiner . . . 43
  - 1.7.4 Welch t test - For all components touched for all commits by Sustained founder(34) against Sustained later joiner . . . 44
- 1.8 Anova generate - For all components touched for all commits . . . 45
  - 1.8.1 Anova generate - For all components touched for a all commits excluding tranient . . . 46
- 1.9 Strategy logistic regression - For all components touched for all commits . . . 48
- 1.10 Tukey hsd - For all components touched for all commits . . . 51

**2 Repository: 14** **53**
- 2.1 Time series sustained components touched by developer - 14 For developer commits and components touched. . . . 53
- 2.2 Time series sustained components touched by developer - 14 For developer day and components touched. . . . 56

**3 Repository: 21** **58**
- 3.1 Time series sustained components touched by developer - 21 For developer commits and components touched. . . . 58
- 3.2 Time series sustained components touched by developer - 21 For developer day and components touched. . . . 61

# 1 Repository

## 1.1 Glossary - Summary

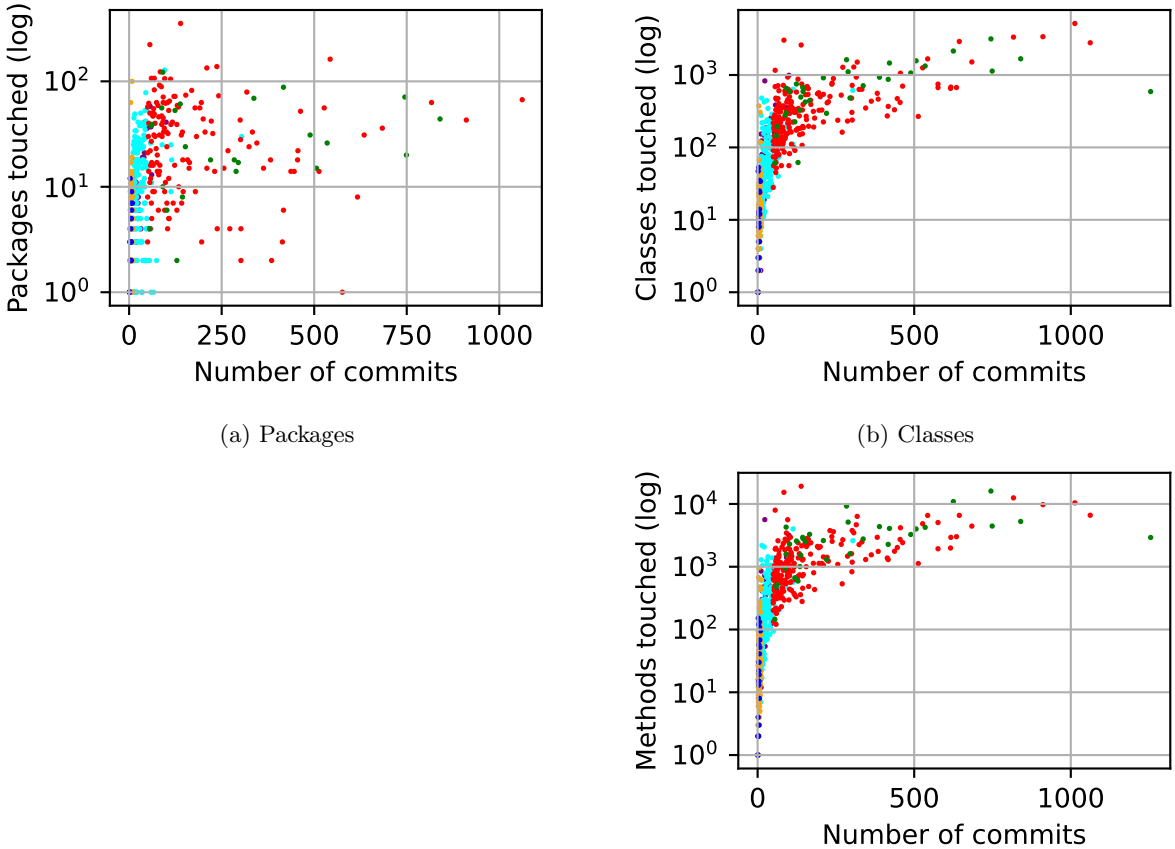
- The founder developer starts in the first six months of a project.
- Late joiner developers begin after six months.
- Transient developers have ten (10) or fewer commits.
- Moderate developers have more than ten (10) commits but fewer than 50 commits or commits for less than 250 days.
- Sustained developers make 50 or more commits and commit for 250 days or more.

## 1.2 Repository Summary Table

Table 1: Summary of fifteen open-source repositories identified from GitHub that have at least 1000 pull requests and at least three sustained late joiner developers. Sustained later joiner developers joining the project after six months and contributed at least 50 commits over a period of 250 days or more. Please note that five developers each worked on two repositories.

| ID | Repo Name | Transient Founder | Moderate Founder | Sustained Founder | Transient Joiner | Moderate Joiner | Sustained Joiner | Commit | Start | End |
|---|---|---|---|---|---|---|---|---|---|---|
| 3 | activiti | 4 | 7 | 6 | 47 | 20 | 5 | 8662 | 2010-Jun-18 | On going |
| 5 | airbyte-platform | 3 | 2 | 1 | 45 | 32 | 6 | 6784 | 2020-Aug-04 | 2023-Sep-27 |
| 7 | ambari | 0 | 3 | 0 | 34 | 32 | 15 | 4247 | 2011-Sep-22 | On going |
| 10 | automq | 2 | 0 | 0 | 40 | 10 | 3 | 1874 | 2011-Aug-10 | 2017-Apr-21 |
| 14 | buck | 8 | 5 | 6 | 119 | 85 | 21 | 14876 | 2013-Mar-21 | 2022-May-05 |
| 15 | camel | 0 | 4 | 1 | 257 | 63 | 13 | 36528 | 2007-Mar-19 | On going |
| 18 | checkstyle | 0 | 0 | 1 | 57 | 42 | 9 | 17338 | 2001-Jun-22 | On going |
| 20 | cxf | 6 | 6 | 3 | 27 | 17 | 3 | 7893 | 2008-Apr-29 | On going |
| 21 | intellij-community | 0 | 3 | 7 | 113 | 93 | 84 | 101247 | 2004-Nov-11 | 2018-Apr-18 |
| 26 | guava | 0 | 1 | 0 | 47 | 8 | 3 | 9685 | 2009-Sep-15 | On going |
| 28 | jenkins | 1 | 1 | 1 | 144 | 39 | 4 | 30252 | 2006-Nov-05 | On going |
| 33 | openmrs-core | 0 | 1 | 0 | 123 | 26 | 3 | 15757 | 2006-May-03 | On going |
| 36 | presto | 0 | 1 | 3 | 162 | 71 | 18 | 30605 | 2012-Aug-09 | On going |
| 37 | quarkus | 1 | 3 | 5 | 126 | 25 | 8 | 81612 | 2018-Jun-14 | On going |
| 39 | selenium | 3 | 3 | 0 | 55 | 17 | 8 | 23354 | 2004-Nov-03 | On going |
| | Total | 28 | 40 | 34 | 1396 | 580 | 203 | 390714 | | |

### 1.3 Sample scatter touched against commit -




(a) Packages

(b) Classes

(c) Methods

Figure 1: Scatter plots of the log total number of components touched (y-axis) against the number of commits (x-axis) made by samples of developers capped at 203 from six (6) categories: Transient Founder (Blue, 28), Moderate Founder (Purple, 40), Sustained Founder (Green, 34), Transient Later Joiner (Orange, 203), Moderate Later Joiner (Cyan, 203), Sustained Later Joiner (Red, 203).

### 1.4 Total commits by developer - Developer Commits

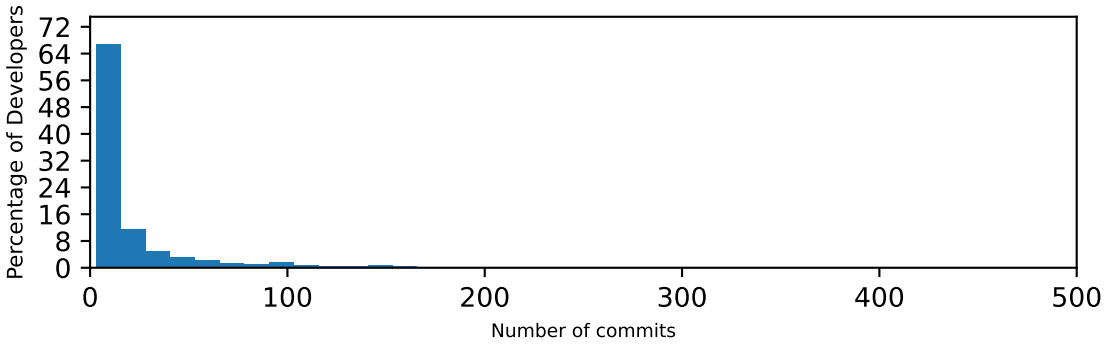
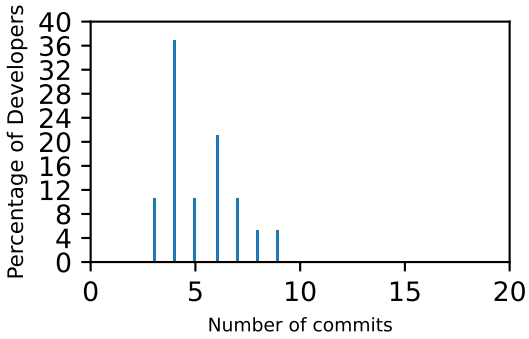
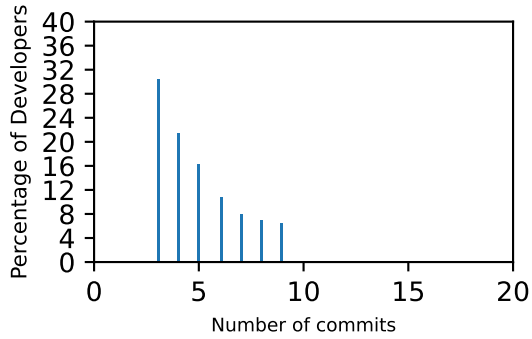


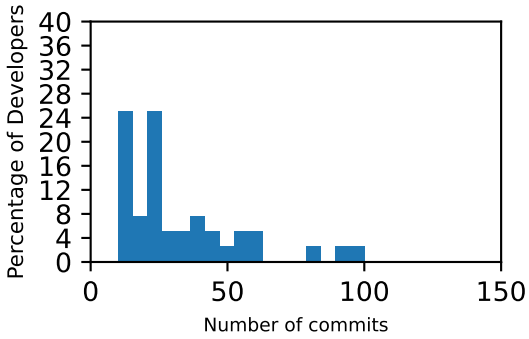

Figure 2: Histogram of the number from all developers (n=1906) (y-axis) against the total number of commits (x-axis) in 15 repositories sampled from GitHub.



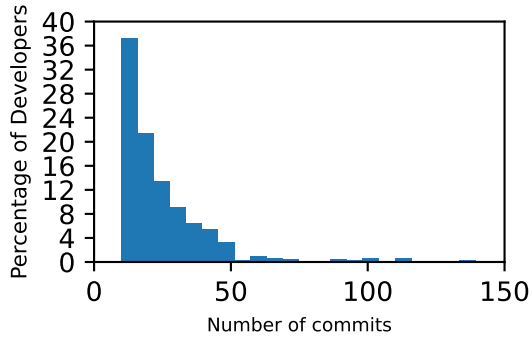

(a) Transient founder (n=19).



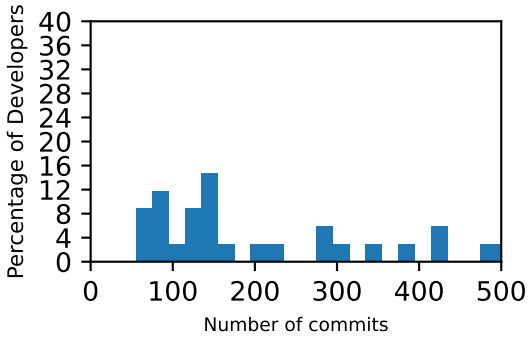

(b) Transient later joiner (n=1030).



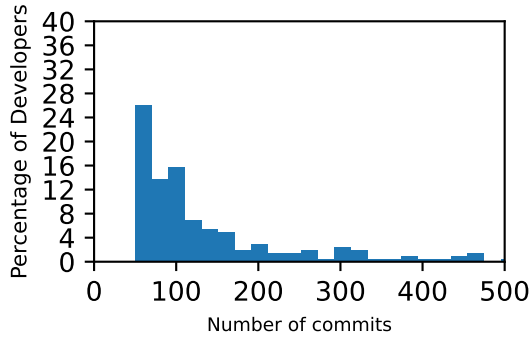

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).



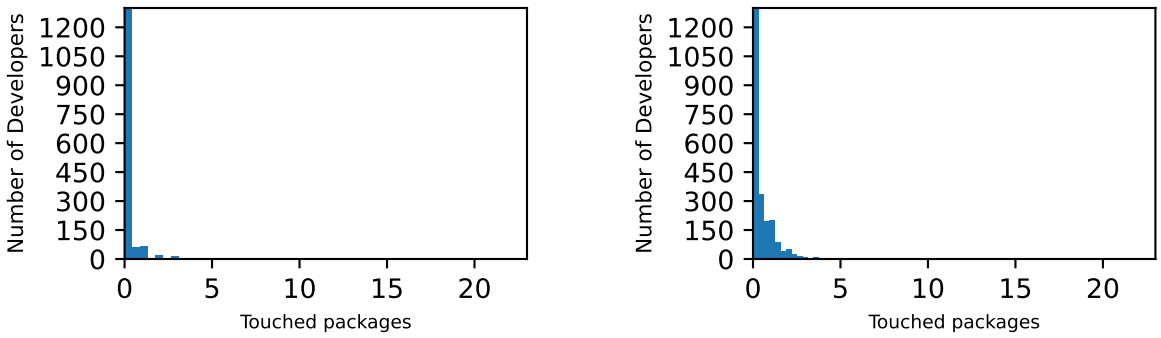

(e) Sustained founder (n=34).




(f) Sustained later joiner (n=203).

Figure 3: Histogram of percentage of developers (y-axis) against total commits made (x-axis) in six (6) categories. This is from 15 repositories sampled from GitHub.

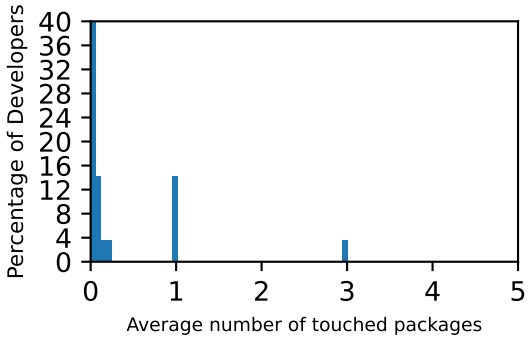
### 1.5 Histogram components touched by developer - For components with total



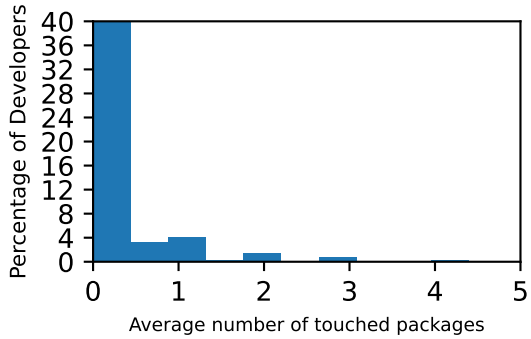

(a) By Day.

(b) By Commit.

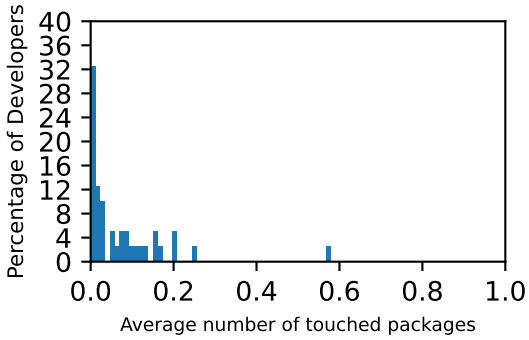
Figure 4: Histogram of the number of developers (y-axis) against the average packages touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



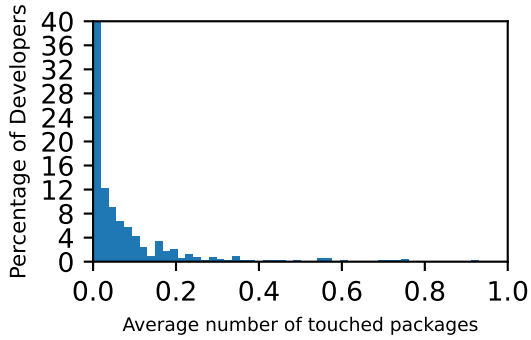

(a) Transient founder (n=28).



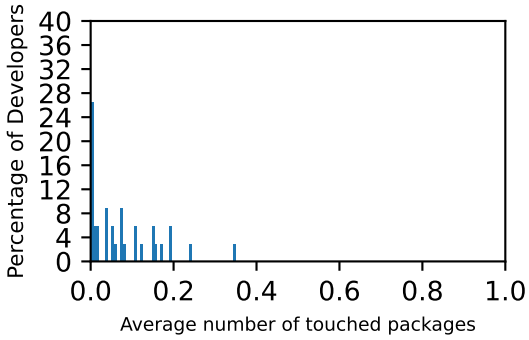

(b) Transient later joiner (n=1396).



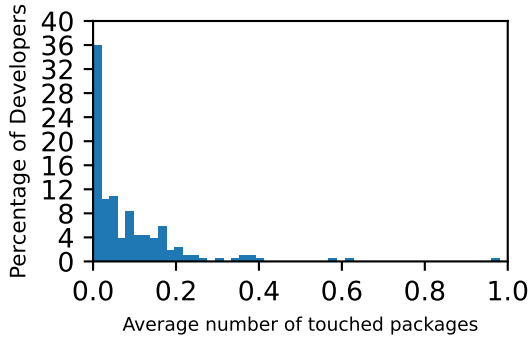

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).

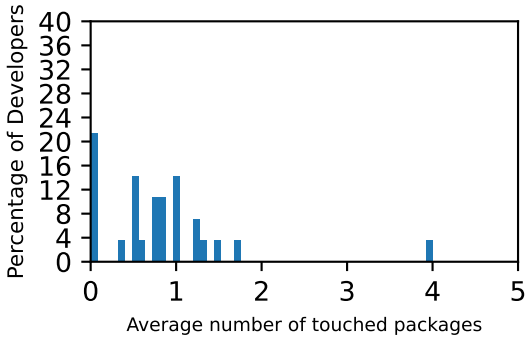



(e) Sustained founder (n=34).

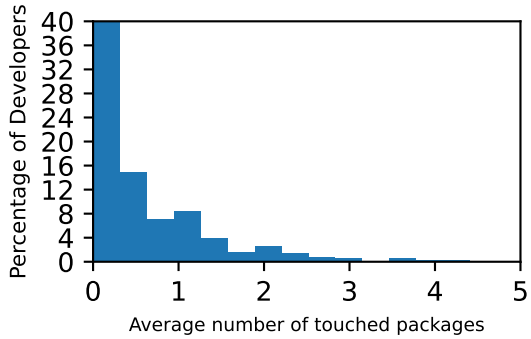



(f) Sustained later joiner (n=203).

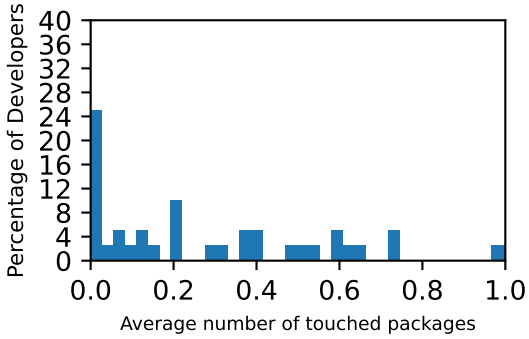
Figure 5: Histogram of the number of developers (y-axis) against the average packages touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



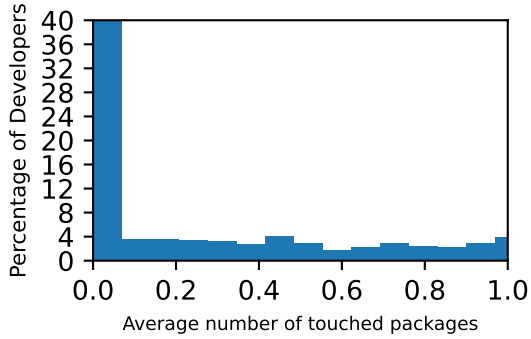

(a) Transient founder (n=28).



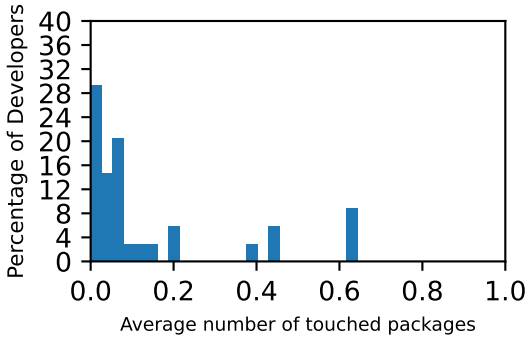

(b) Transient later joiner (n=1396).



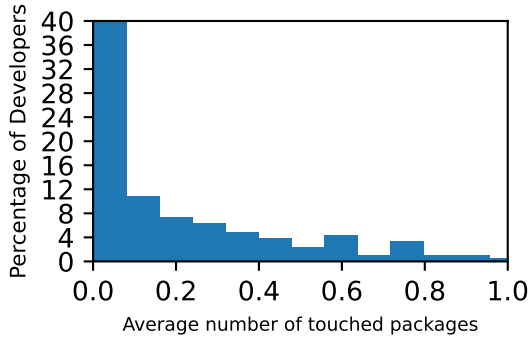

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).

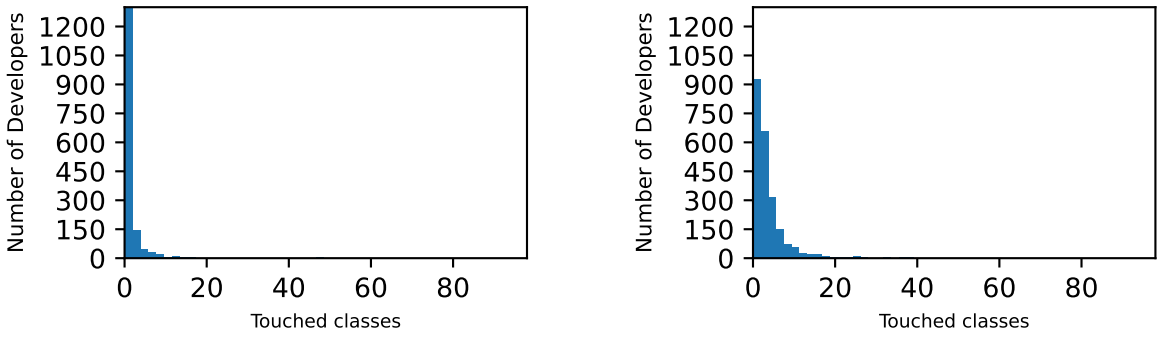



(e) Sustained founder (n=34).




(f) Sustained later joiner (n=203).

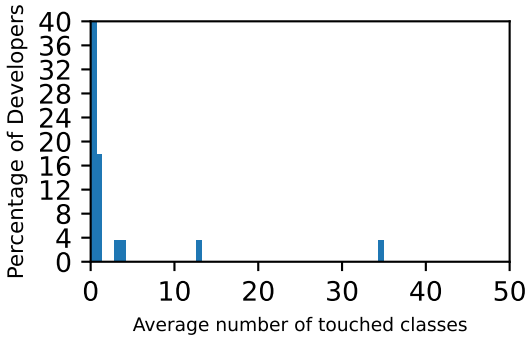
Figure 6: Histogram of the number of developers (y-axis) against the average packages touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



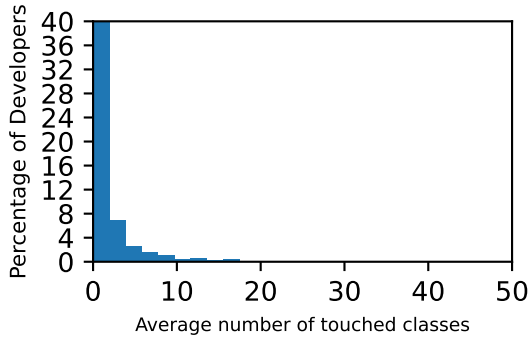

(a) By Day.

(b) By Commit.

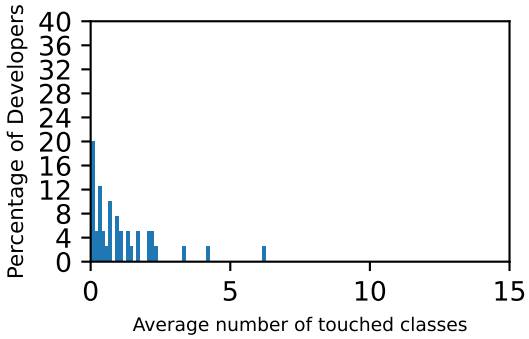
Figure 7: Histogram of the number of developers (y-axis) against the average classes touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



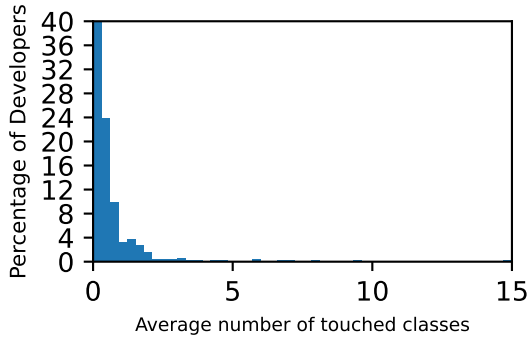

(a) Transient founder (n=28).



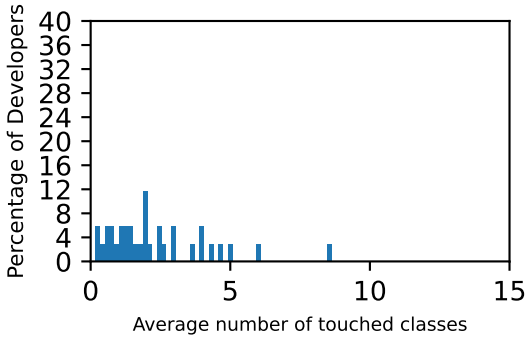

(b) Transient later joiner (n=1396).



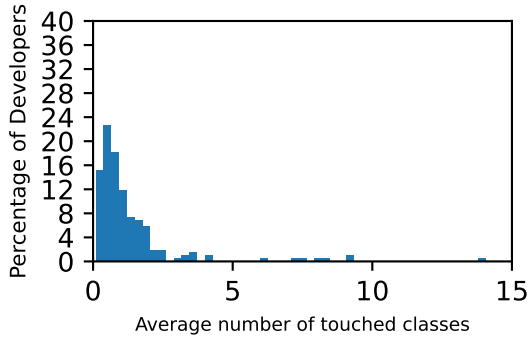

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).

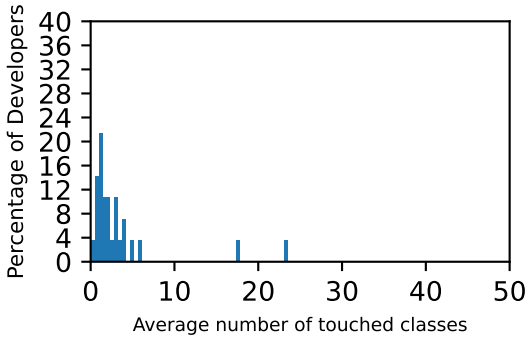



(e) Sustained founder (n=34).

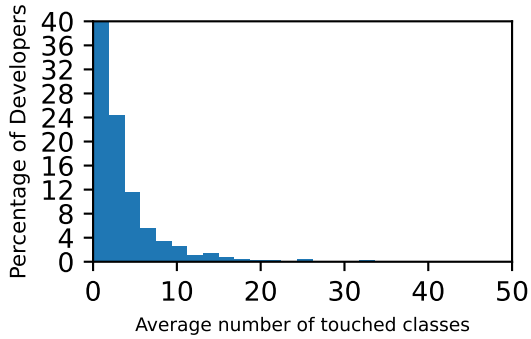



(f) Sustained later joiner (n=203).

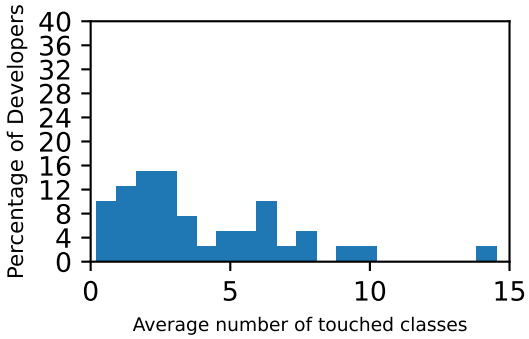
Figure 8: Histogram of the number of developers (y-axis) against the average classes touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



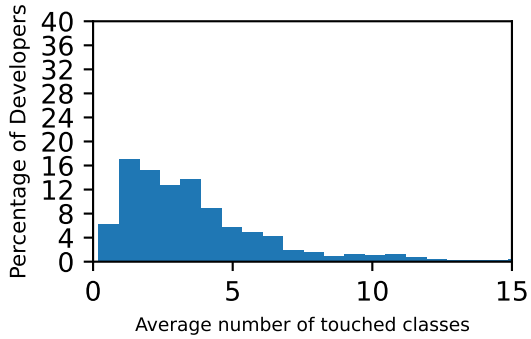

(a) Transient founder (n=28).



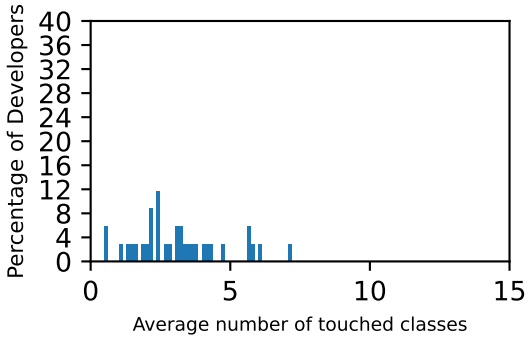

(b) Transient later joiner (n=1396).



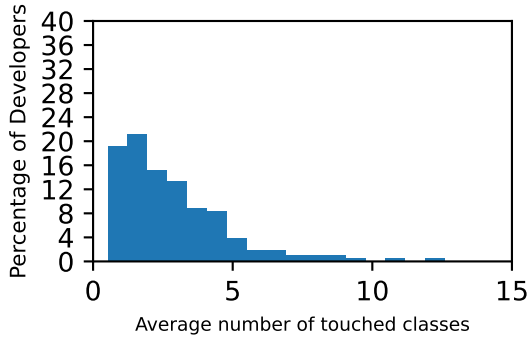

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).

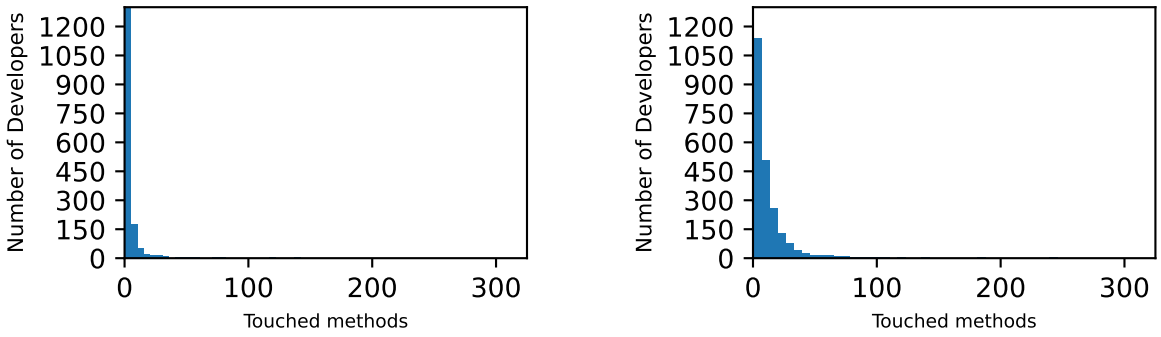



(e) Sustained founder (n=34).




(f) Sustained later joiner (n=203).

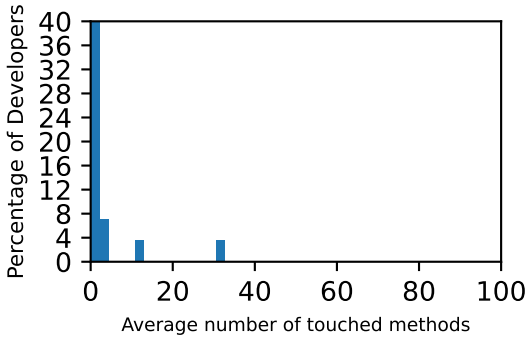
Figure 9: Histogram of the number of developers (y-axis) against the average classes touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



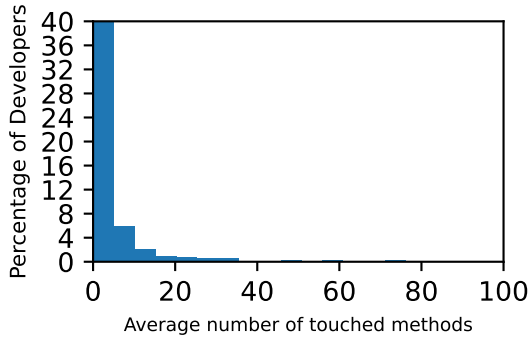

(a) By Day.

(b) By Commit.

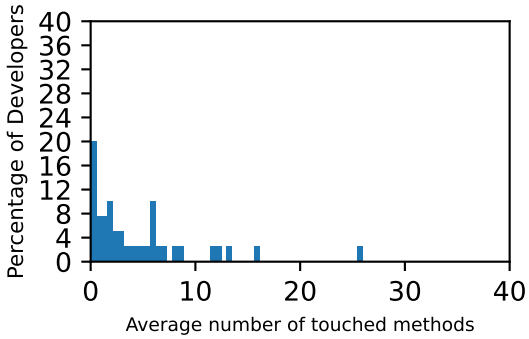
Figure 10: Histogram of the number of developers (y-axis) against the average methods touched by day and commit (x-axis) from all 2278 developers from 15 repositories sampled from GitHub.



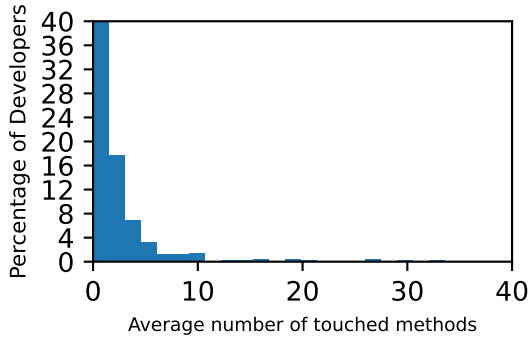

(a) Transient founder (n=28).



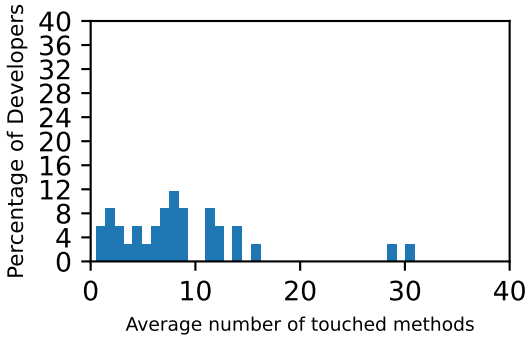

(b) Transient later joiner (n=1396).



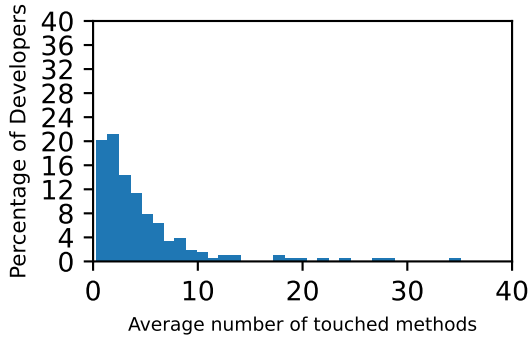

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).

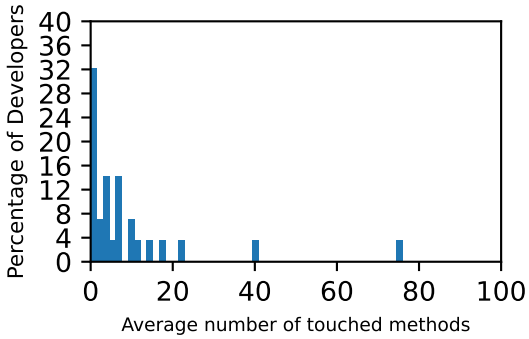



(e) Sustained founder (n=34).

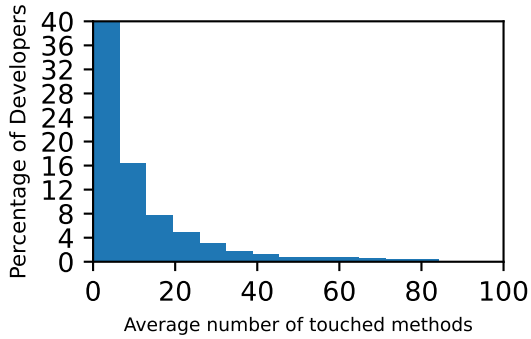



(f) Sustained later joiner (n=203).

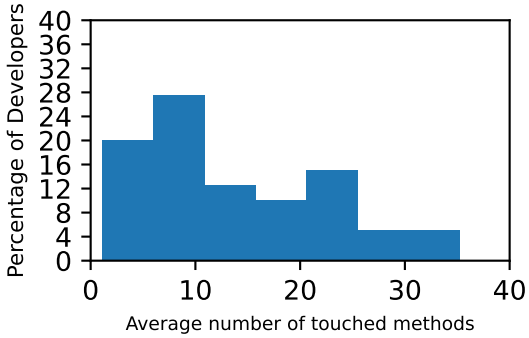
Figure 11: Histogram of the number of developers (y-axis) against the average methods touched by day (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.



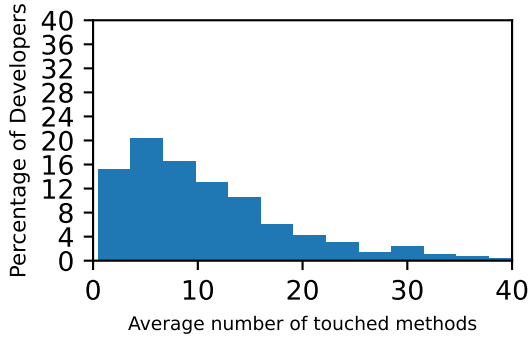

(a) Transient founder (n=28).



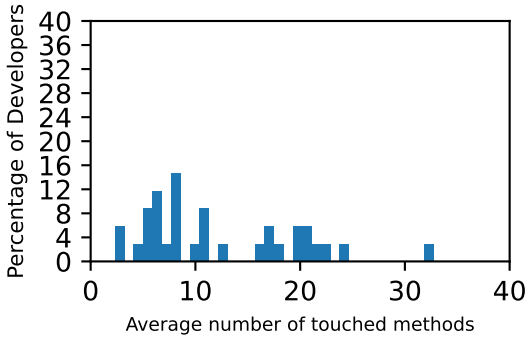

(b) Transient later joiner (n=1396).



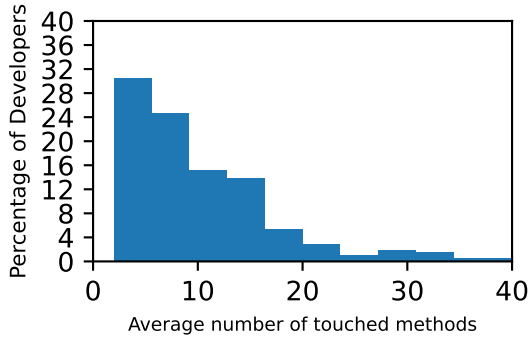

(c) Moderate founder (n=40).




(d) Moderate later joiner (n=580).




(e) Sustained founder (n=34).

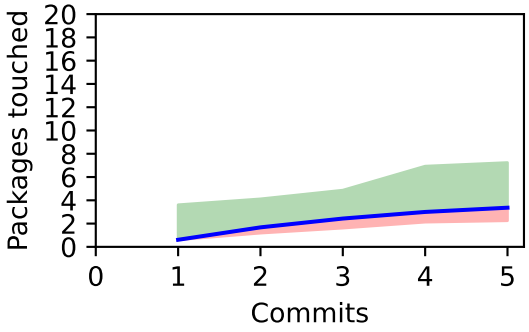



(f) Sustained later joiner (n=203).

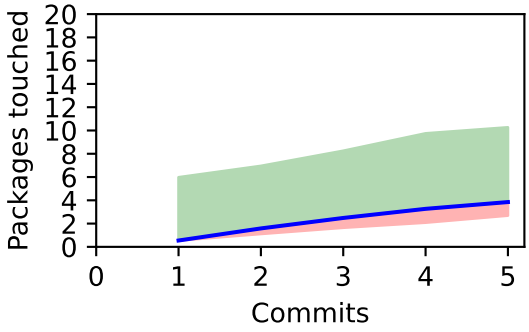
Figure 12: Histogram of the number of developers (y-axis) against the average methods touched by commit (x-axis) from six (6) categories of developers from 15 repositories sampled from GitHub.

## 1.6 Time series components touched - For all components touched for all commits and for each period

### 1.6.1 Time series components touched - For packages touched for first five commits

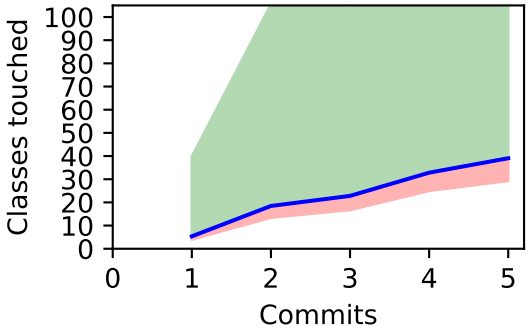



(a) Packages for All founders (n=74 & $\mu$=3.36), excluding transient

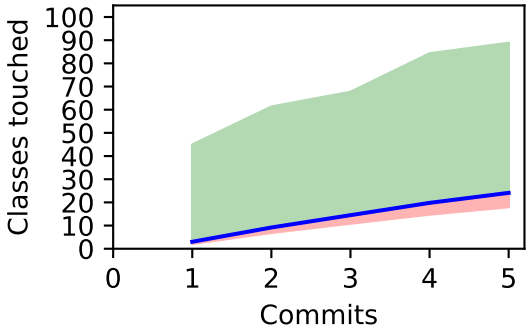



(b) Packages for All joiners (n=783 & $\mu$=3.85), excluding transient

Figure 13: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

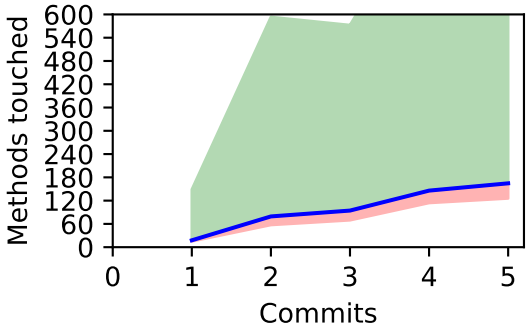



(a) Classes for All founders (n=74 & $\mu$=39.05), excluding transient

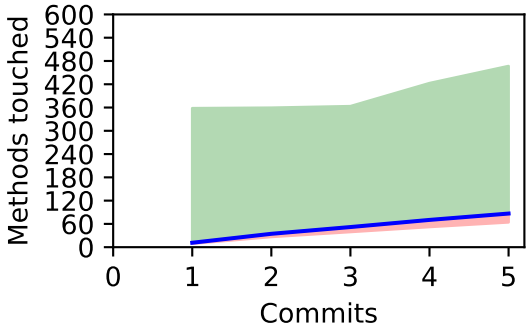



(b) Classes for All joiners (n=783 & $\mu$=24.11), excluding transient

Figure 14: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

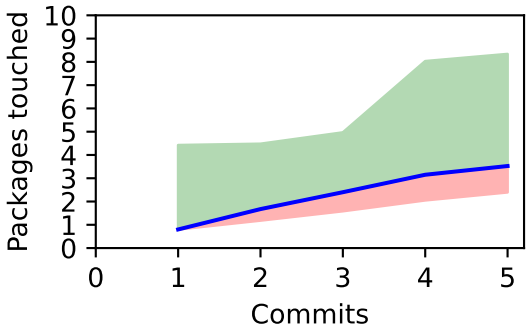



(a) Methods for All founders (n=74 & $\mu$=164.59), excluding transient

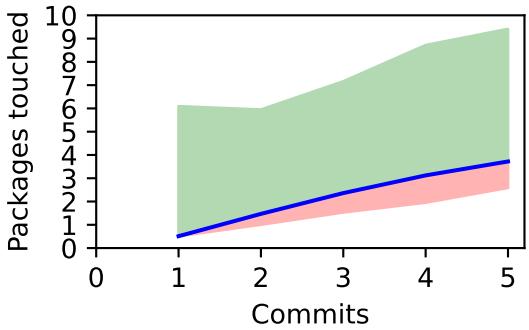



(b) Methods for All joiners (n=783 & $\mu$=86.68), excluding transient

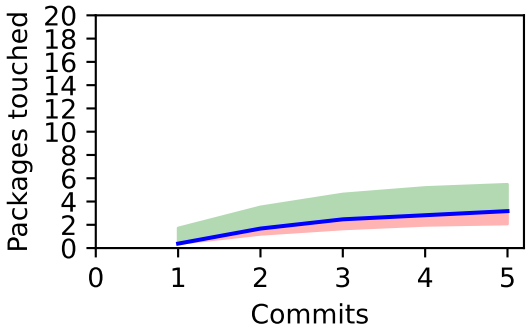
Figure 15: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.



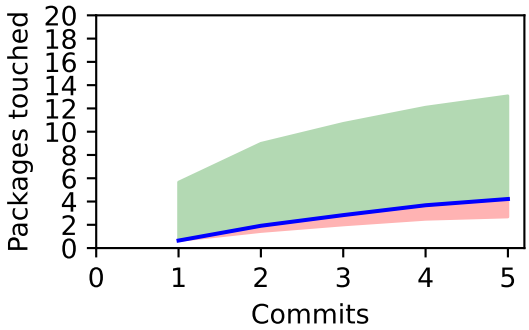

(a) Moderate founder (n=40 & $\mu$=3.52)




(b) Moderate later joiner (n=580 & $\mu$=3.72)

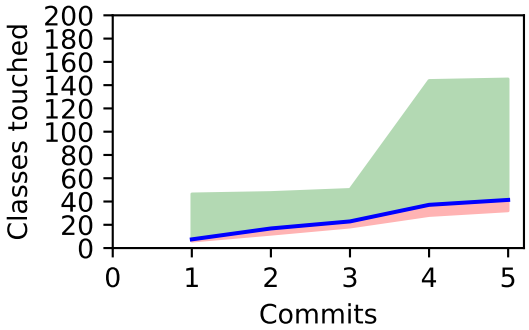



(c) Sustained founder (n=34 & $\mu$=3.18)

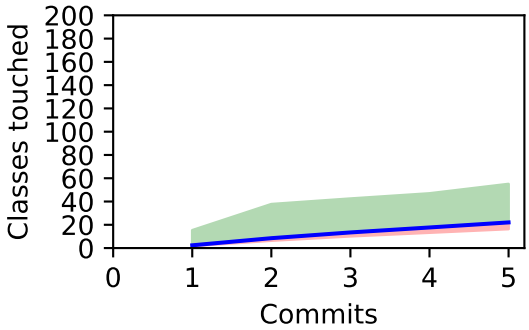



(d) Sustained later joiner (n=203 & $\mu$=4.22)

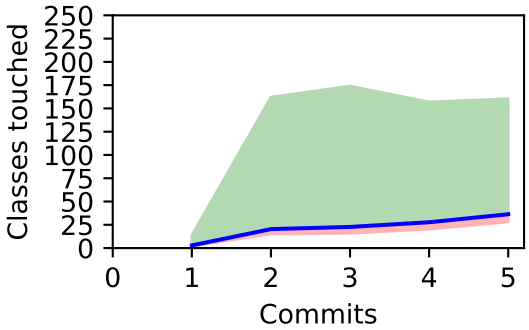
Figure 16: A time series of the average mean (blue line) total packages touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.



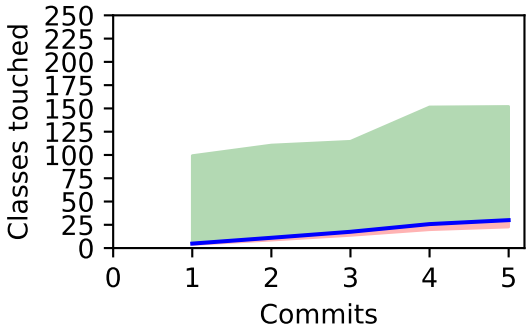

(a) Moderate founder (n=40 & $\mu$=41.38)




(b) Moderate later joiner (n=580 & $\mu$=22.04)

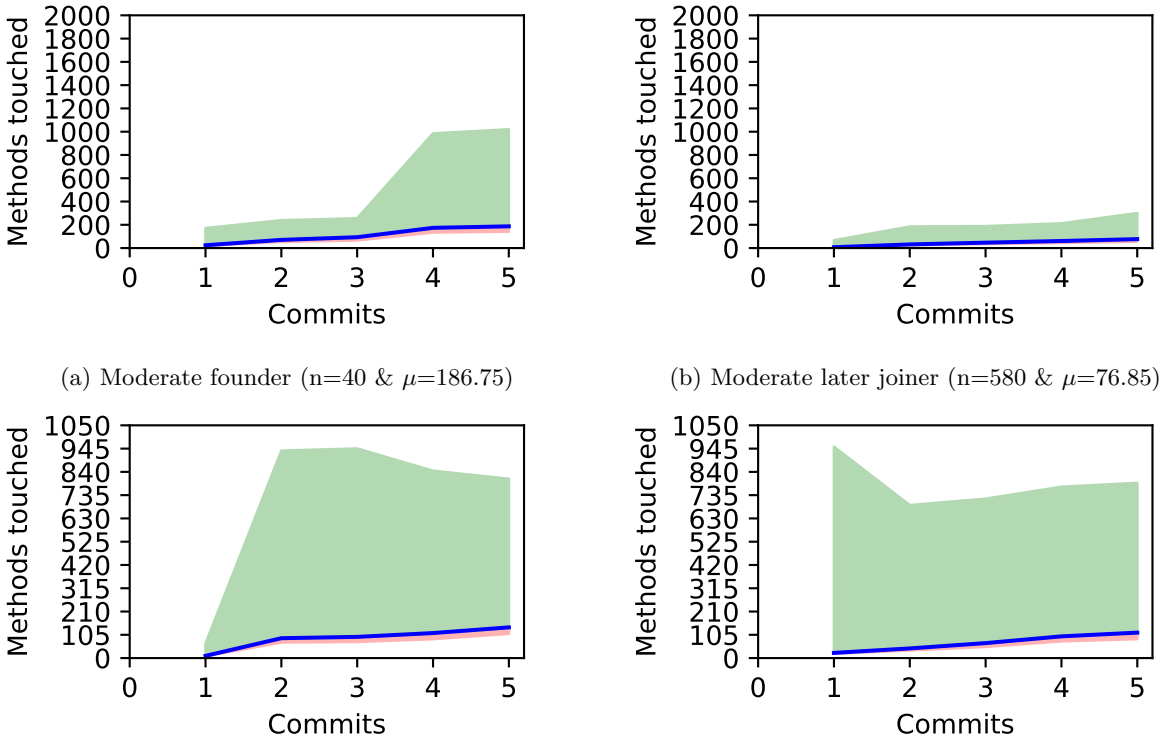



(c) Sustained founder (n=34 & $\mu$=36.32)




(d) Sustained later joiner (n=203 & $\mu$=30.01)

Figure 17: A time series of the average mean (blue line) total classes touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.




(a) Moderate founder (n=40 & $\mu$=186.75)

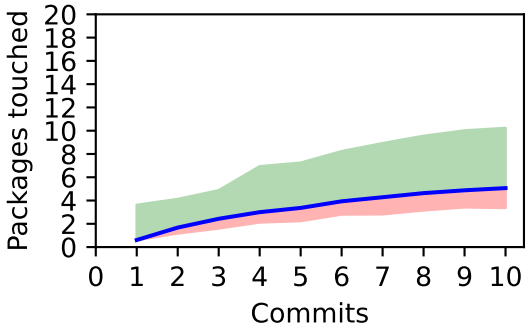
(b) Moderate later joiner (n=580 & $\mu$=76.85)

(c) Sustained founder (n=34 & $\mu$=138.53)

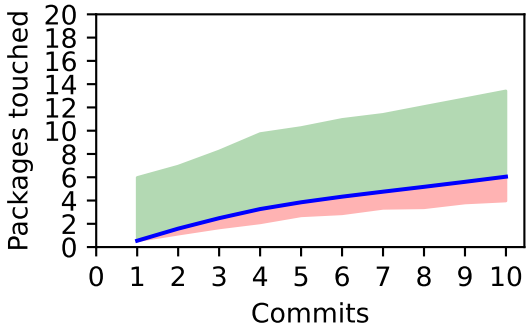
(d) Sustained later joiner (n=203 & $\mu$=114.77)

Figure 18: A time series of the average mean (blue line) total methods touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.

### 1.6.2 Time series components touched - For packages touched for first ten commits

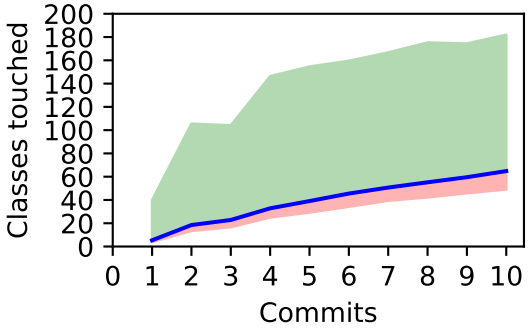



(a) Packages for All founders (n=74 & $\mu$=5.07), excluding transient

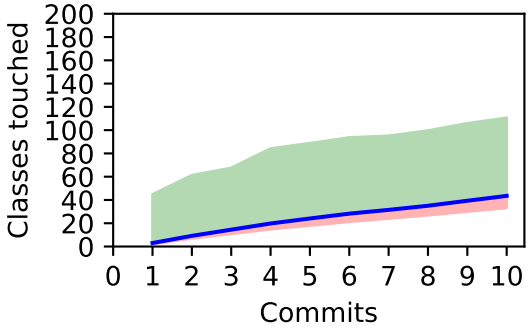



(b) Packages for All joiners (n=783 & $\mu$=6.05), excluding transient

Figure 19: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

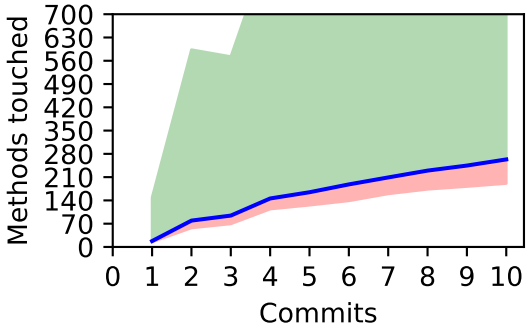



(a) Classes for All founders (n=74 & $\mu$=64.81), excluding transient

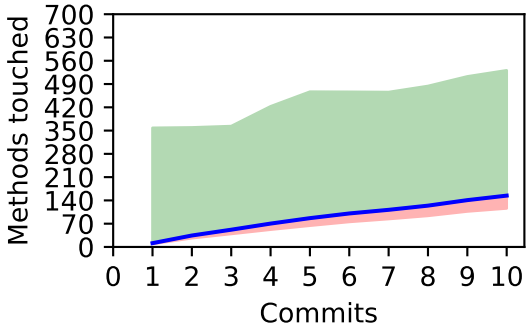



(b) Classes for All joiners (n=783 & $\mu$=43.51), excluding transient

Figure 20: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

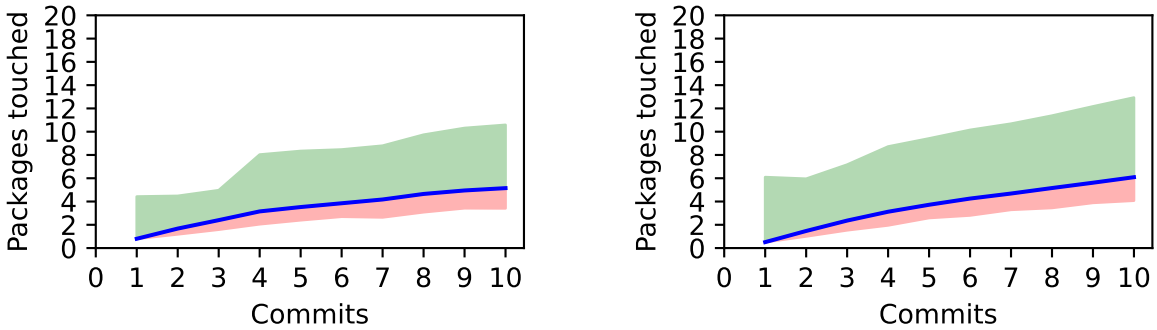



(a) Methods for All founders (n=74 & $\mu$=263.26), excluding transient




(b) Methods for All joiners (n=783 & $\mu$=154.41), excluding transient

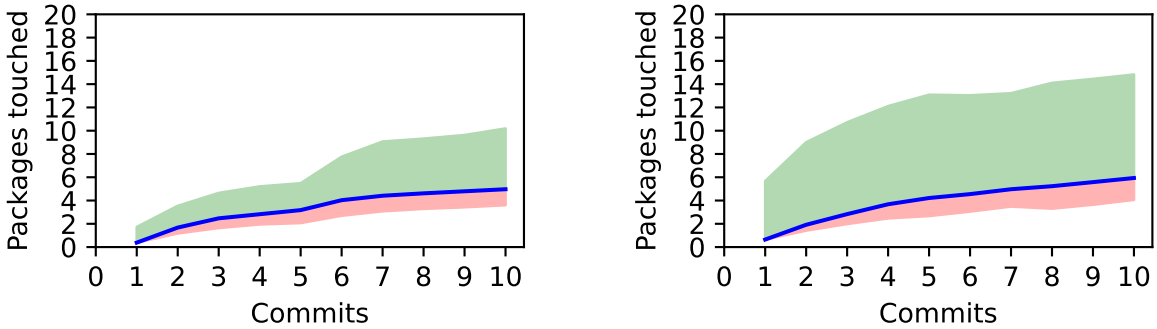
Figure 21: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.




(a) Moderate founder (n=40 & $\mu$=5.15)




(b) Moderate later joiner (n=580 & $\mu$=6.09)

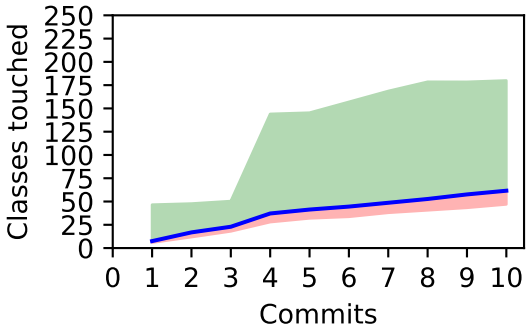



(c) Sustained founder (n=34 & $\mu$=4.97)

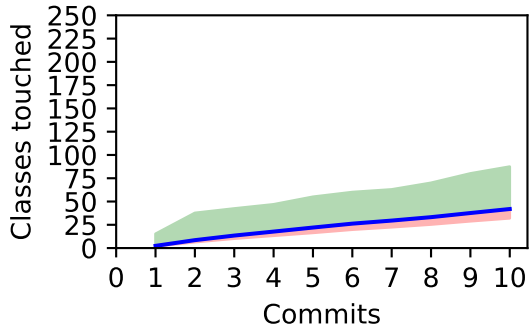



(d) Sustained later joiner (n=203 & $\mu$=5.94)

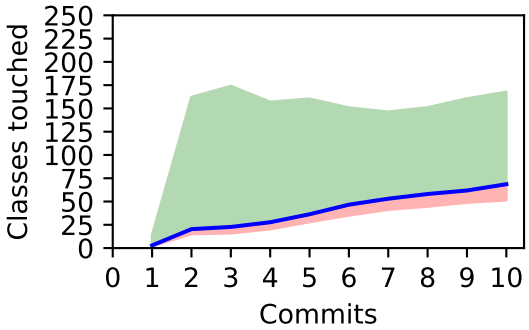
Figure 22: A time series of the average mean (blue line) total packages touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.



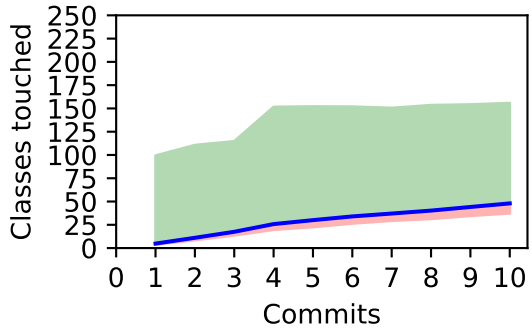

(a) Moderate founder (n=40 & $\mu$=61.6)




(b) Moderate later joiner (n=580 & $\mu$=41.95)

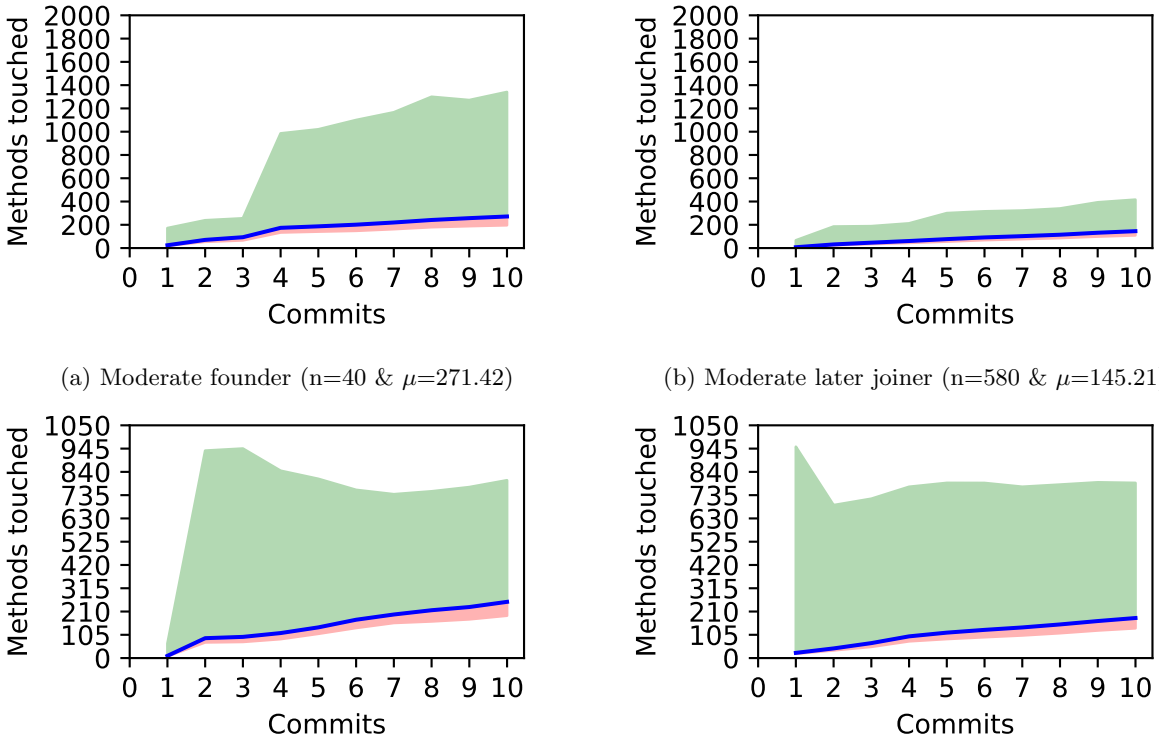



(c) Sustained founder (n=34 & $\mu$=68.59)




(d) Sustained later joiner (n=203 & $\mu$=47.98)

Figure 23: A time series of the average mean (blue line) total classes touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.




(a) Moderate founder (n=40 & $\mu$=271.42)

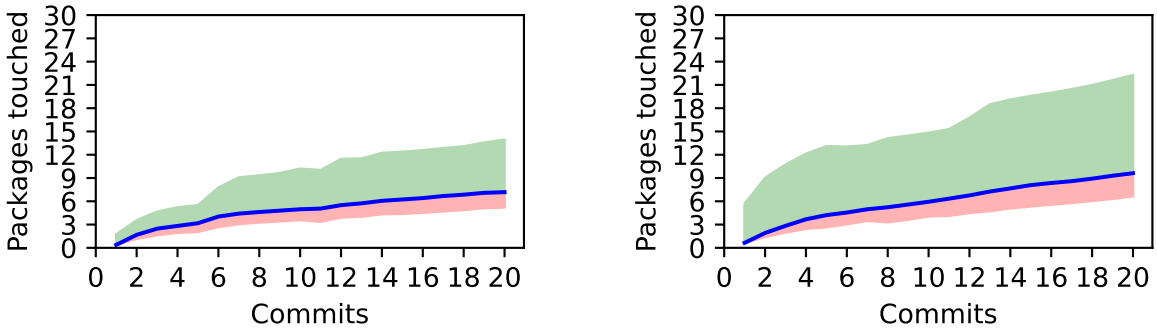
(b) Moderate later joiner (n=580 & $\mu$=145.21)

(c) Sustained founder (n=34 & $\mu$=253.65)

(d) Sustained later joiner (n=203 & $\mu$=180.7)

Figure 24: A time series of the average mean (blue line) total methods touched (y-axis) against the number of commits (x-axis) for four (4) categories of developer. Green is std above mean and pink std below mean.

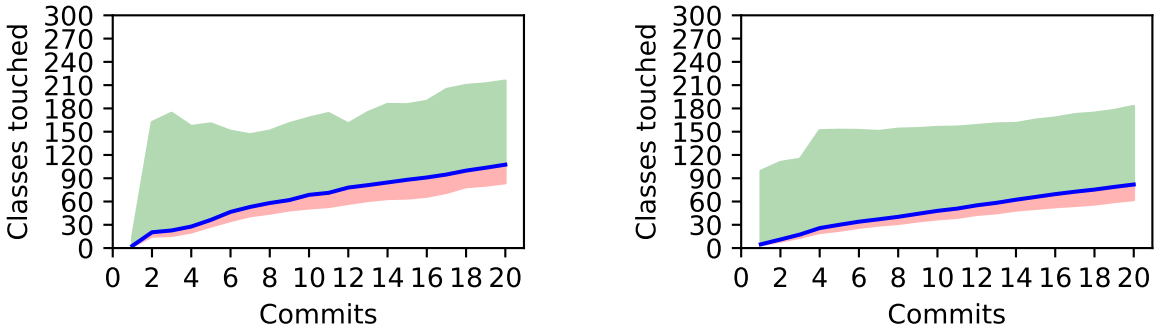
### 1.6.3 Time series components touched - For packages touched for first twenty commits




(a) Sustained founder (n=34 & $\mu$=7.18)

(b) Sustained later joiner (n=203 & $\mu$=9.62)

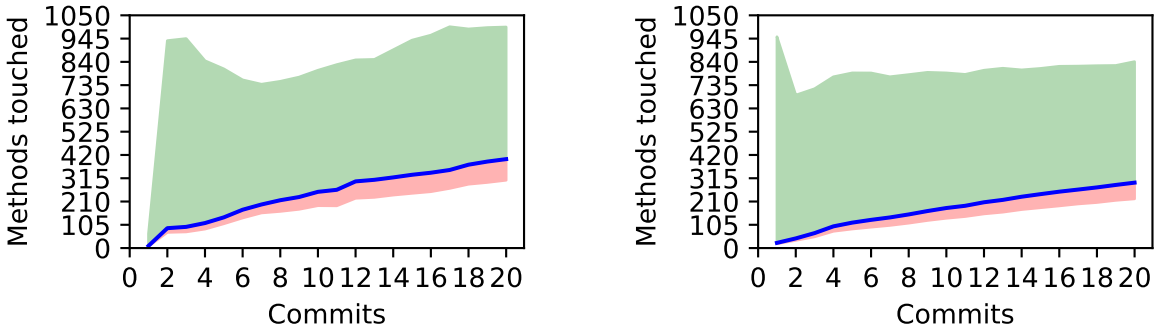
Figure 25: A time series of the average mean (blue line) total packages touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.




(a) Sustained founder (n=34 & $\mu$=107.5)

(b) Sustained later joiner (n=203 & $\mu$=81.91)

Figure 26: A time series of the average mean (blue line) total classes touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.

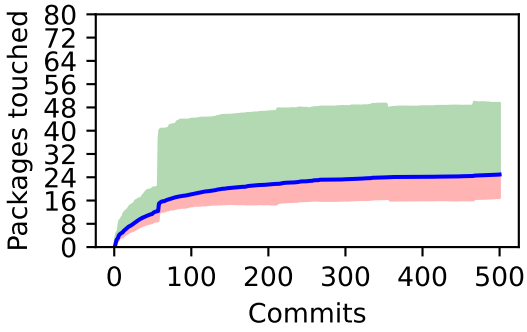



(a) Sustained founder (n=34 & $\mu$=401.24)

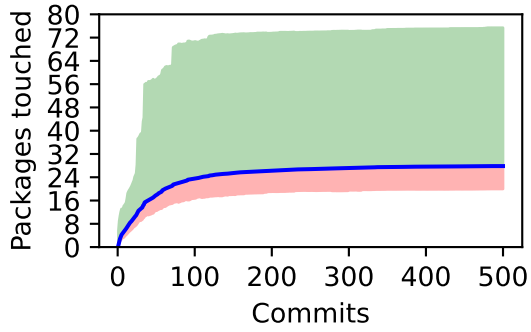
(b) Sustained later joiner (n=203 & $\mu$=295.1)

Figure 27: A time series of the average mean (blue line) total methods touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.

#### 1.6.4 Time series components touched - For packages touched for commit

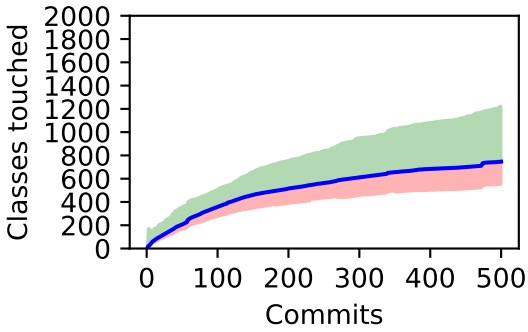



(a) Packages for All founders (n=34 & $\mu$=24.94), excluding transient

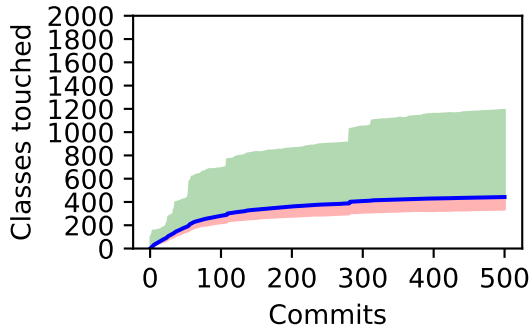



(b) Packages for All joiners (n=203 & $\mu$=27.84), excluding transient

Figure 28: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

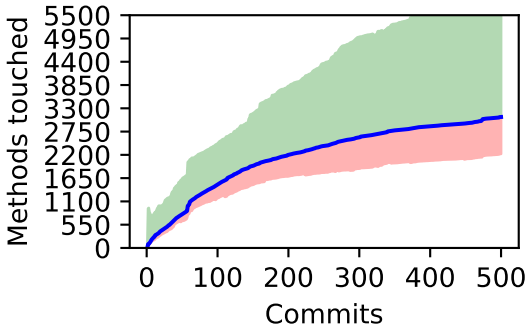



(a) Classes for All founders (n=34 & $\mu$=746.59), excluding transient

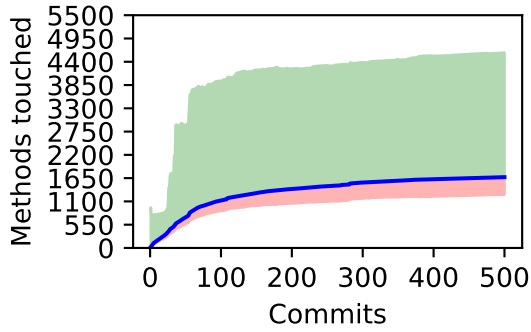



(b) Classes for All joiners (n=203 & $\mu$=442.33), excluding transient

Figure 29: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

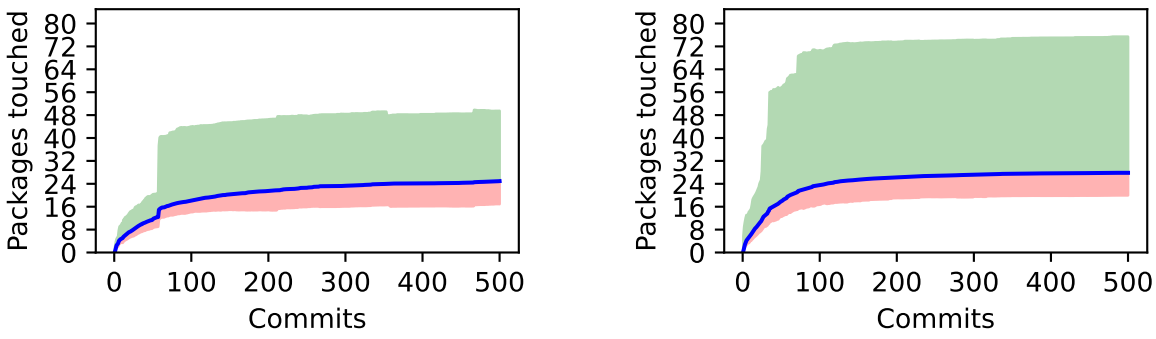



(a) Methods for All founders (n=34 & $\mu$=3095.18), excluding transient




(b) Methods for All joiners (n=203 & $\mu$=1672.84), excluding transient

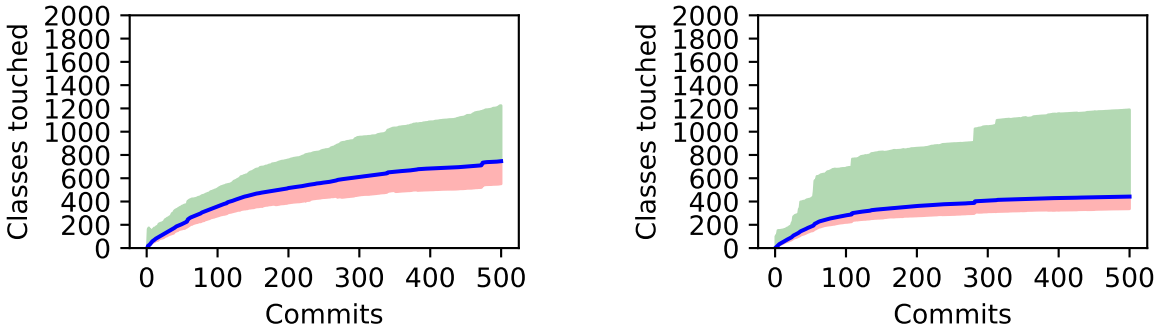
Figure 30: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of commits (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.




(a) Sustained founder (n=34 & $\mu$=24.94)

(b) Sustained later joiner (n=203 & $\mu$=27.84)

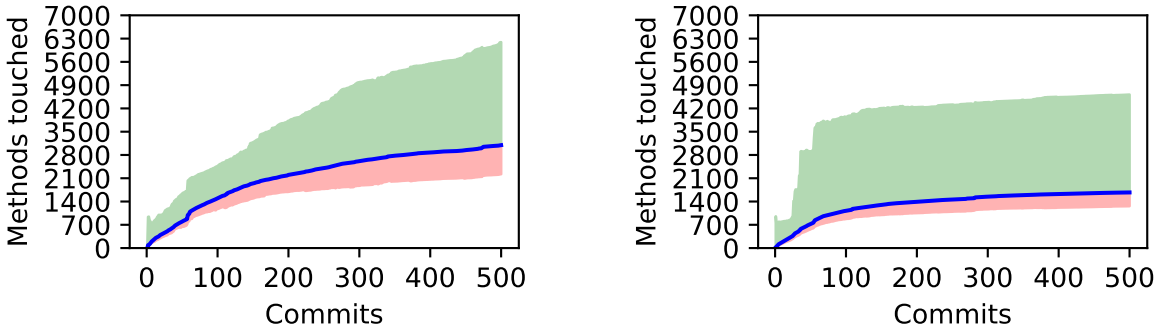
Figure 31: A time series of the average mean (blue line) total packages touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.




(a) Sustained founder (n=34 & $\mu$=746.59)

(b) Sustained later joiner (n=203 & $\mu$=442.33)

Figure 32: A time series of the average mean (blue line) total classes touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.

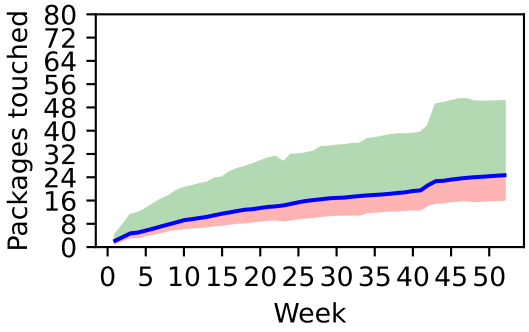



(a) Sustained founder (n=34 & $\mu$=3095.18)

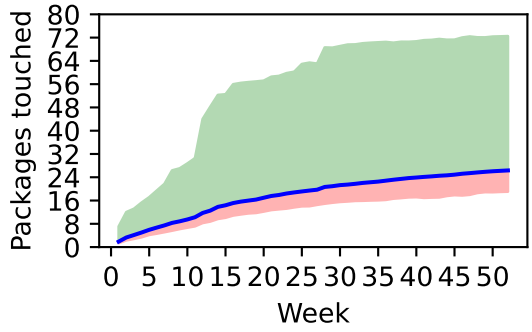
(b) Sustained later joiner (n=203 & $\mu$=1672.84)

Figure 33: A time series of the average mean (blue line) total methods touched (y-axis) against the number of commits (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.

#### 1.6.5 Time series components touched - For packages touched for each week commit

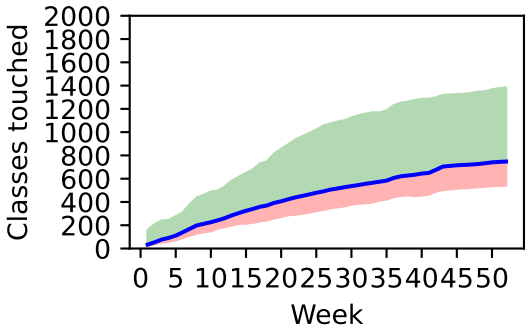



(a) Packages for All founders (n=34 & $\mu$=24.7), excluding transient

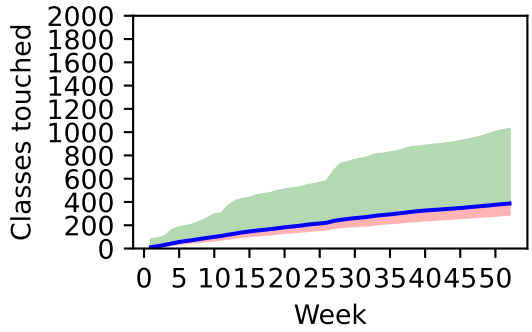



(b) Packages for All joiners (n=203 & $\mu$=26.35), excluding transient

Figure 34: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of week (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

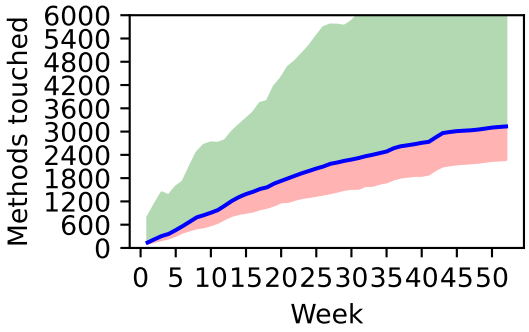



(a) Classes for All founders (n=34 & $\mu$=747.76), excluding transient

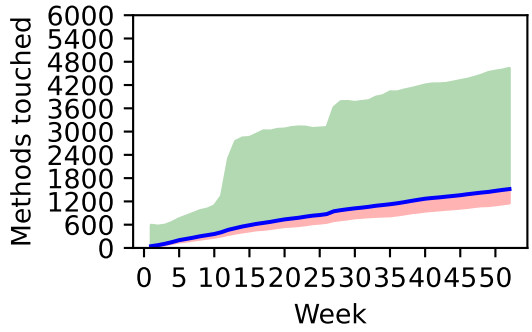



(b) Classes for All joiners (n=203 & $\mu$=386.64), excluding transient

Figure 35: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of week (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.

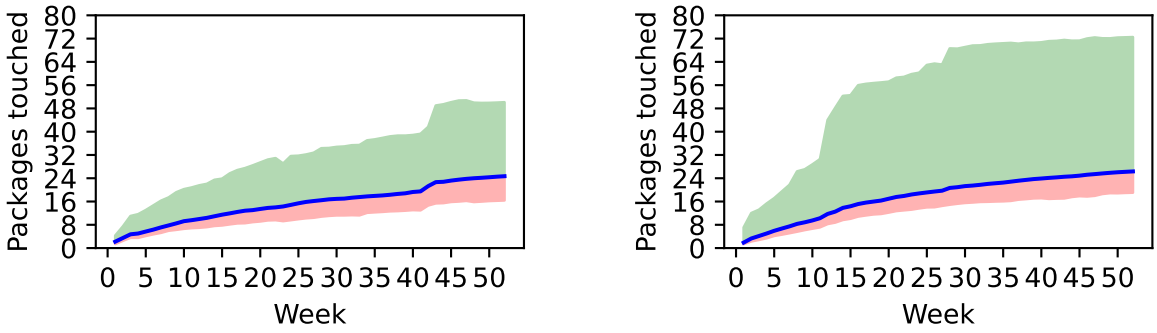



(a) Methods for All founders (n=34 & $\mu$=3132.41), excluding transient




(b) Methods for All joiners (n=203 & $\mu$=1518.36), excluding transient

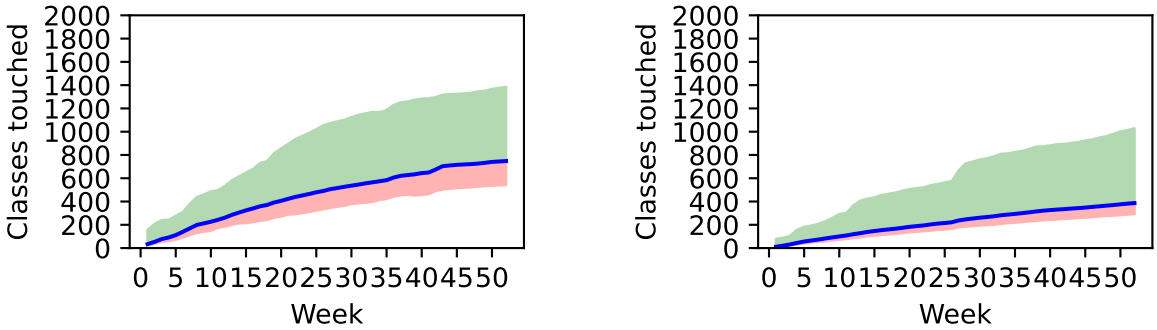
Figure 36: A time series of the average mean (blue line) total for all components touched (y-axis) against the number of week (x-axis) for developers. Green is std variation above mean and pink is std variation below mean.




(a) Sustained founder (n=34 & $\mu$=24.7)

(b) Sustained later joiner (n=203 & $\mu$=26.35)

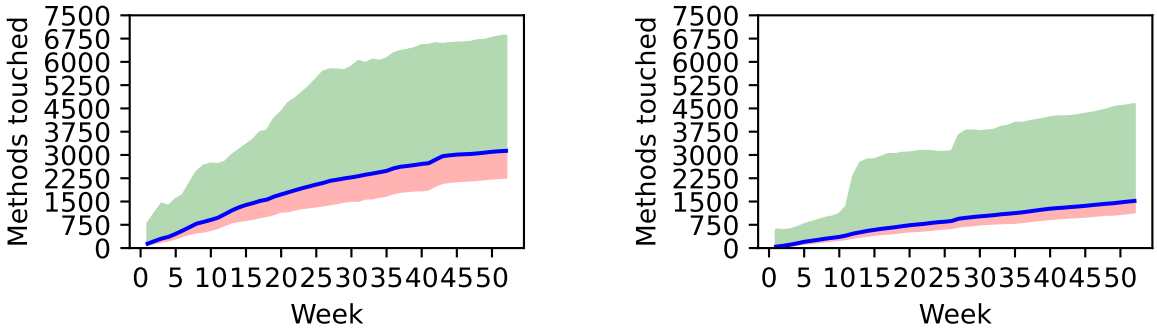
Figure 37: A time series of the average mean (blue line) total packages touched (y-axis) against the number of week (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.




(a) Sustained founder (n=34 & $\mu$=747.76)

(b) Sustained later joiner (n=203 & $\mu$=386.64)

Figure 38: A time series of the average mean (blue line) total classes touched (y-axis) against the number of week (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.




(a) Sustained founder (n=34 & $\mu$=3132.41)

(b) Sustained later joiner (n=203 & $\mu$=1518.36)

Figure 39: A time series of the average mean (blue line) total methods touched (y-axis) against the number of week (x-axis) for two (2) categories of developer. Green is std above mean and pink std below mean.

## 1.7 Welch t test - For all components touched for all commits

A Welch $t$-test was employed to determine if the mean contributions of Group A differ significantly from Group B. This test is appropriate for continuous data and is robust to unequal variances. Complementarily, the Mann-Whitney $U$ (MWU) test was used to assess stochastic dominance by ranking data points. This non-parametric approach does not require a normal distribution and is highly resistant to outliers. Columns below:

- **Component** - Packages, classes or method touched.
- **Number of first commits** - A sample of the first x number of commits for developers.
- **Sample A** $\mu$ - The mean average of the first sample A.
- **Sample B** $\mu$ - The mean average of the second sample B.
- **Welch Statisic** - This is the magnitude of the change, greater than 2.0 generally indicates a significant change.
- **Welch** $P$ - A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.
- **MWU Statsitcal (Number of Pairs)** - The Mann-Whitney $U$ Statistic represents the number of pairwise comparisons where one group outranks the other, with the total possible pairs provided in brackets.
- **MWU** $P$ - The Mann-Whitney $p$-value. A significance test of the difference between sample A and sample B. A $p < 0.05$ indicates a significant difference between samples.

### 1.7.1 Welch t test - For all components touched for all commits by Founder(74) against Joiner (783)

Table 2: Welch t-test and Mann-Whitney U results for components and number of commits by founder (74) against joiner (783)

| Categories | Component | Number of First Commits | Founder $\mu$ | Joiner $\mu$ | Welch Statistic | Welch $P$ | MWU Statistic (Number of Pairs) | MWU $P$ |
|---|---|---|---|---|---|---|---|---|
| moderate & sustained | Packages | 5 | 3.36 | 3.85 | -0.99 | 0.3232 | 29,616.5 (57,942) | 0.7459 |
| moderate & sustained | Packages | 10 | 5.07 | 6.05 | -1.33 | 0.1855 | 28,593.0 (57,942) | 0.8498 |
| moderate & sustained | Classes | 5 | 39.05 | 24.11 | 1.77 | 0.0799 | 32,451.0 (57,942) | 0.0872 |
| moderate & sustained | Classes | 10 | 64.81 | 43.51 | 2.07 | 0.0419 | 32,846.5 (57,942) | 0.0569 |
| moderate & sustained | Methods | 5 | 164.59 | 86.68 | 1.56 | 0.1218 | 32,324.5 (57,942) | 0.0995 |
| moderate & sustained | Methods | 10 | 263.26 | 154.41 | 1.76 | 0.0826 | 33,245.0 (57,942) | 0.0358 |

### 1.7.2 Welch t test - For all components touched for all commits by Moderate(620) against Sustained (237)

Table 3: Welch t-test and Mann-Whitney U results for components and number of commits by moderate (620) against sustained (237)

| Categories | Component | Number of First Commits | Moderate $\mu$ | Sustained $\mu$ | Welch Statistic | Welch $P$ | MWU Statistic (Number of Pairs) | MWU $P$ |
|---|---|---|---|---|---|---|---|---|
| founder & later joiner | Packages | 5 | 3.71 | 4.07 | -0.80 | 0.4217 | 70,136.5 (146,940) | 0.2931 |
| founder & later joiner | Packages | 10 | 6.03 | 5.80 | 0.40 | 0.6905 | 73,098.0 (146,940) | 0.9068 |
| founder & later joiner | Classes | 5 | 23.29 | 30.92 | -1.69 | 0.0919 | 74,229.5 (146,940) | 0.8148 |
| founder & later joiner | Classes | 10 | 43.22 | 50.94 | -1.51 | 0.1316 | 74,941.0 (146,940) | 0.6500 |
| founder & later joiner | Methods | 5 | 83.94 | 118.18 | -1.48 | 0.1389 | 73,502.5 (146,940) | 0.9921 |
| founder & later joiner | Methods | 10 | 153.35 | 191.16 | -1.49 | 0.1366 | 73,214.0 (146,940) | 0.9372 |

### 1.7.3 Welch t test - For all components touched for all commits by Moderate founder(40) against Moderate later joiner (580)

Table 4: Welch t-test and Mann-Whitney U results for components and number of commits by moderate founder (40) against moderate later joiner (580)

| Categories | Component | Number of First Commits | Moderate founder $\mu$ | Moderate later joiner $\mu$ | Welch Statistic | Welch $P$ | MWU Statistic (Number of Pairs) | MWU $P$ |
|---|---|---|---|---|---|---|---|---|
| | Packages | 5 | 3.52 | 3.72 | -0.27 | 0.7874 | 12,075.0 (23,200) | 0.6567 |
| | Packages | 10 | 5.15 | 6.09 | -0.95 | 0.3462 | 11,610.5 (23,200) | 0.9925 |
| | Classes | 5 | 41.38 | 22.04 | 1.70 | 0.0969 | 13,640.0 (23,200) | 0.0626 |
| | Classes | 10 | 61.60 | 41.95 | 1.40 | 0.1679 | 13,313.0 (23,200) | 0.1180 |
| | Methods | 5 | 186.75 | 76.85 | 1.46 | 0.1516 | 13,748.5 (23,200) | 0.0499 |
| | Methods | 10 | 271.43 | 145.21 | 1.28 | 0.2063 | 13,417.5 (23,200) | 0.0973 |

### 1.7.4 Welch t test - For all components touched for all commits by Sustained founder(34) against Sustained later joiner (203)

Table 5: Welch t-test and Mann-Whitney U results for components and number of commits by sustained founder (34) against sustained later joiner (203)

| Categories | Component | Number of First Commits | Sustained founder $\mu$ | Sustained later joiner $\mu$ | Welch Statistic | Welch $P$ | MWU Statistic (Number of Pairs) | MWU $P$ |
|---|---|---|---|---|---|---|---|---|
| | Packages | 5 | 3.18 | 4.22 | -1.47 | 0.1459 | 3,369.0 (6,902) | 0.8231 |
| | Packages | 10 | 4.97 | 5.94 | -0.83 | 0.4084 | 3,298.5 (6,902) | 0.6772 |
| | Packages | 20 | 7.18 | 9.62 | -1.41 | 0.1628 | 3,181.5 (6,902) | 0.4616 |
| | Classes | 5 | 36.32 | 30.01 | 0.48 | 0.6350 | 3,651.0 (6,902) | 0.5895 |
| | Classes | 10 | 68.59 | 47.98 | 1.30 | 0.2003 | 3,879.0 (6,902) | 0.2478 |
| | Classes | 20 | 107.50 | 81.91 | 1.34 | 0.1863 | 3,846.5 (6,902) | 0.2857 |
| | Methods | 5 | 138.53 | 114.77 | 0.36 | 0.7211 | 3,550.5 (6,902) | 0.7890 |
| | Methods | 10 | 253.65 | 180.70 | 1.00 | 0.3240 | 3,896.5 (6,902) | 0.2291 |
| | Methods | 20 | 401.24 | 295.10 | 1.16 | 0.2533 | 3,693.5 (6,902) | 0.5131 |

## 1.8 Anova generate - For all components touched for all commits

An Anova table for all components touched on average.

The Two-Way ANOVA evaluates the influence of two independent categorical variables—*Role* (Founder vs. Late Joiner) and *Tenure* (Moderate vs. Sustained)—on the number of unique architectural components touched. The analysis excludes and includes Transient developers to isolate the behaviors of contributors with established project engagement.

**Explanation of Table Columns:**

- **Sum Sq (Sum of Squares):** Represents the total variation explained by that specific factor. A higher Sum Sq relative to the Residual indicates that the factor has a strong influence on the outcome.
- **df (Degrees of Freedom):** The number of independent values that can vary in the calculation. For categorical variables, this is typically $N - 1$ levels.
- **F (F-statistic):** The ratio of the variance explained by the factor to the variance unexplained (Residual). A larger F-value suggests the factor is more likely to be significant.
- **p-Value ($PR(> F)$):** The probability that the observed differences occurred by chance. Using a significance threshold of $\alpha = 0.05$, values above this indicate no statistically significant effect.
- **ETA Sq** The effect size, quantifies the proportion of the total variance in your dependent variable.

**Explanation of Table Rows:**

- **Intercept:** Represents the baseline value of the dependent variable when all categorical factors are at their reference levels.
- **Role(F/L):** Tests the *Main Role Effect* of being a Founder versus a Late Joiner. The p-Value above 0.05 indicates that the timing of project entry does not significantly change the breadth of early-stage component exploration.
- **Category(M/S):** Tests the *Main Category Effect* of the eventual contributor catregory (Transient vs Moderate vs. Sustained).
- **Interaction:** Tests the *Interaction Effect*. It evaluates if the effect of being a Founder is different for the different categories.
- **Residual:** Represents the "noise" or variance within the groups that cannot be explained by Role or Tenure.

### 1.8.1 Anova generate - For all components touched for a all commits excluding tranient

Table 6: Anova results for all components, number of commits and 857 developers excluding transient

| Component | Number of First Commits | Factor | Sum Squared | df | F-Stat | $P$ Value | ETA Sq |
|---|---|---|---|---|---|---|---|
| packages | 5 | Role (F/L) | 1.43 | 1 | 0.05 | 0.8250 | 0.0001 |
| packages | 5 | Category (M/S) | 2.23 | 1 | 0.08 | 0.7826 | 0.0001 |
| packages | 5 | Interaction | 11.68 | 1 | 0.40 | 0.5279 | 0.0005 |
| packages | 5 | Residual | 24984.13 | 853 | - | - | |
| packages | 10 | Role (F/L) | 33.04 | 1 | 0.59 | 0.4443 | 0.0007 |
| packages | 10 | Category (M/S) | 0.59 | 1 | 0.01 | 0.9185 | 0.0000 |
| packages | 10 | Interaction | 0.01 | 1 | 0.00 | 0.9889 | 0.0000 |
| packages | 10 | Residual | 48127.58 | 853 | - | - | |
| classes | 5 | Role (F/L) | 13981.95 | 1 | 7.35 | 0.0069 | 0.0082 |
| classes | 5 | Category (M/S) | 468.97 | 1 | 0.25 | 0.6198 | 0.0003 |
| classes | 5 | Interaction | 2774.71 | 1 | 1.46 | 0.2276 | 0.0016 |
| classes | 5 | Residual | 1623543.63 | 853 | - | - | |
| classes | 10 | Role (F/L) | 14448.45 | 1 | 5.02 | 0.0254 | 0.0055 |
| classes | 10 | Category (M/S) | 897.52 | 1 | 0.31 | 0.5769 | 0.0003 |
| classes | 10 | Interaction | 15.03 | 1 | 0.01 | 0.9424 | 0.0000 |
| classes | 10 | Residual | 2457107.31 | 853 | - | - | |
| methods | 5 | Role (F/L) | 451979.70 | 1 | 8.27 | 0.0041 | 0.0093 |
| methods | 5 | Category (M/S) | 42733.87 | 1 | 0.78 | 0.3769 | 0.0009 |
| methods | 5 | Interaction | 121524.53 | 1 | 2.22 | 0.1364 | 0.0025 |
| methods | 5 | Residual | 46646591.43 | 853 | - | - | |
| methods | 10 | Role (F/L) | 596144.39 | 1 | 7.60 | 0.0060 | 0.0085 |
| methods | 10 | Category (M/S) | 5808.58 | 1 | 0.07 | 0.7856 | 0.0001 |
| methods | 10 | Interaction | 46476.28 | 1 | 0.59 | 0.4417 | 0.0007 |
| methods | 10 | Residual | 66927848.79 | 853 | - | - | |

**Statistical Model**

The differences between developer categories are assessed using a Two-Way ANOVA with interaction. The value for any individual developer metric ($Y_{ijk}$) is modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk} \tag{1}$$

Where:

- $\mu$ is the grand mean of the touched components.
- $\alpha_i$ represents the effect of the **Role** (Founder vs. Late Joiner).
- $\beta_j$ represents the effect of the **Tenure** (Transient, Moderate, Sustained).
- $(\alpha\beta)_{ij}$ is the interaction effect between Role and Tenure.
- $\epsilon_{ijk}$ is the residual error.

The $F$-statistic for each factor is calculated by:

$$F = \frac{SS_{factor}/df_{factor}}{SS_{residual}/df_{residual}} \tag{2}$$

## 1.9 Strategy logistic regression - For all components touched for all commits

To identify early indicators of long-term developer engagement, a Logistic Regression model was employed. The model treats the developer category (Sustained vs. Non-Sustained) as a binary dependent variable. The following metrics are reported:

- **Coefficient ($\beta$):** This value represents the change in the log-odds of a developer becoming 'Sustained' for every unit increase in the component count. A positive coefficient indicates that higher activity in that component increases the likelihood of long-term retention.
- **Odds Ratio ($e^{\beta}$):** Perhaps the most interpretable metric, the Odds Ratio quantifies the relative increase in the likelihood of being a Sustained developer. For example, an Odds Ratio of 1.05 implies a 5% increase in the odds of becoming sustained for each additional component touched.
- **Std. Err (Standard Error):** Measures the accuracy of the coefficient estimate. A lower standard error relative to the coefficient indicates a more precise estimate.
- **p-Value ($P > |z|$):** Indicates the statistical significance of the predictor. In this study, a p-Value $< 0.05$ suggests that the specific component count is a significant predictor of whether a developer will transition into a sustained role.
- **50% Threshold:** The threshold number of components to touch to have a 50
- **Moderate Met Threshold:** The moderate number of developers to have met the threshold, the percentate of developers in brackets.
- **Sustained Met Threshold:** The sustained number of developers to have met the threshold, the percentate of developers in brackets.

**Interpretation of Intercept vs. Component Count:**

- **Intercept:** Represents the baseline log-odds of the dependent variable (Sustained Status) when the independent variable (e.g., Package Count) is zero. In this study, the consistently negative intercepts indicate that the probability of a developer becoming 'Sustained' is statistically low without active engagement in the codebase.
- **Component Count (Slope):** Represents the effect size of architectural engagement. The positive coefficients and Odds Ratios greater than 1.0 demonstrate that every additional unique component touched by a developer significantly increases their likelihood of transitioning into a sustained contributor.

Table 7: Strategy Logistic Regression for all components against number of commits for 857 developers excluding trainsient

| Component | Number of First Commits | Type | Moderate $\mu$ | Sustained $\mu$ | Coefficient | Odds Ratio | Std. Err | $p$-Value | 30% Threshold | Moderate Met Threshold | Sustained Met Threshold |
|---|---|---|---|---|---|---|---|---|---|---|---|
| packages | 5 | Intercept | 7.47 | 16.48 | -1.2933 | 0.2744 | 0.0975 | 0.0000 | - | - | - |
| packages | 5 | Packages Count | 7.47 | 16.48 | 0.0313 | 1.0318 | 0.0054 | 0.0000 | **14.2540** | 115 (18.5%) | 78 (32.9%) |
| packages | 10 | Intercept | 9.82 | 23.71 | -1.4111 | 0.2439 | 0.1019 | 0.0000 | - | - | - |
| packages | 10 | Packages Count | 9.82 | 23.71 | 0.0306 | 1.0311 | 0.0043 | 0.0000 | **18.4307** | 123 (19.8%) | 89 (37.6%) |
| classes | 5 | Classes Count | 56.44 | 214.75 | 0.0095 | 1.0096 | 0.0010 | 0.0000 | **110.3711** | 86 (13.9%) | 128 (54.0%) |
| classes | 10 | Intercept | 79.92 | 381.41 | -2.6261 | 0.0724 | 0.1577 | 0.0000 | - | - | - |
| classes | 10 | Classes Count | 79.92 | 381.41 | 0.0110 | 1.0111 | 0.0009 | 0.0000 | **161.6419** | 68 (11.0%) | 160 (67.5%) |
| methods | 5 | Methods Count | 207.07 | 878.82 | 0.0019 | 1.0019 | 0.0002 | 0.0000 | **431.2120** | 66 (10.6%) | 115 (48.5%) |
| methods | 10 | Intercept | 290.47 | 1515.71 | -2.2454 | 0.1059 | 0.1368 | 0.0000 | - | - | - |
| methods | 10 | Methods Count | 290.47 | 1515.71 | 0.0023 | 1.0023 | 0.0002 | 0.0000 | **615.7727** | 66 (10.6%) | 151 (63.7%) |

**Predictive Identification Model**

To determine if early-stage developer behavior can identify future sustained contributors, we employ a Logistic Regression model. The probability $P$ of a developer becoming a **Sustained Contributor** is modeled using the log-odds (logit) function:

$$\ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \epsilon \tag{3}$$

Where:

- $P$ is the probability that the developer's tenure is "Sustained".
- $\beta_0$ is the intercept.
- $X_n$ represents early-stage metrics (e.g., methods or classes touched in the first 10, 20, or 50 commits).
- $\beta_n$ are the coefficients representing the weight of each predictor.

The impact of each behavior is quantified using the **Odds Ratio (OR)**, calculated as:

$$OR = e^{\beta_n} \tag{4}$$

An $OR > 1$ indicates that an increase in the metric (e.g., touching more classes) increases the likelihood of long-term retention.

**Model Evaluation** The predictive power is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). An AUC of 0.5 represents random chance, while an AUC closer to 1.0 indicates high predictive accuracy.

## 1.10 Tukey hsd - For all components touched for all commits

An Tukey Honestly Significant Difference table for all components touched on average.

**Post-hoc Analysis (Tukey HSD):** Following the ANOVA, a Tukey Honestly Significant Difference (HSD) test was performed. While ANOVA identifies if at least one group differs, Tukey HSD performs pairwise comparisons to identify specific differences between developer roles and tenure categories while controlling for Type I error rates across multiple comparisons.

Table 8: Tukey HSD results for classes and number of commits, for 857 developers

| Component | Commits | Group A | Group B | Mean Diff | Adj $P$ Value |
|---|---|---|---|---|---|
| classes | 5 | Founder Moderate | Founder Sustained | -5.05 | 0.9599 |
| classes | 5 | Founder Moderate | Late Joiner Moderate | -19.33 | 0.0345 |
| classes | 5 | Founder Moderate | Late Joiner Sustained | -11.37 | 0.4344 |
| classes | 5 | Founder Sustained | Late Joiner Moderate | -14.28 | 0.2486 |
| classes | 5 | Founder Sustained | Late Joiner Sustained | -6.31 | 0.8631 |
| classes | 5 | Late Joiner Moderate | Late Joiner Sustained | 7.96 | 0.1137 |
| classes | 10 | Founder Moderate | Founder Sustained | 6.99 | 0.9443 |
| classes | 10 | Founder Moderate | Late Joiner Moderate | -19.65 | 0.1135 |
| classes | 10 | Founder Moderate | Late Joiner Sustained | -13.62 | 0.4581 |
| classes | 10 | Founder Sustained | Late Joiner Moderate | -26.64 | 0.0258 |
| classes | 10 | Founder Sustained | Late Joiner Sustained | -20.61 | 0.1631 |
| classes | 10 | Late Joiner Moderate | Late Joiner Sustained | 6.03 | 0.5137 |
| classes | 20 | Founder Sustained | Late Joiner Sustained | -25.59 | 0.1034 |
| methods | 5 | Founder Moderate | Founder Sustained | -48.22 | 0.8132 |
| methods | 5 | Founder Moderate | Late Joiner Moderate | -109.90 | 0.0215 |
| methods | 5 | Founder Moderate | Late Joiner Sustained | -71.98 | 0.2840 |
| methods | 5 | Founder Sustained | Late Joiner Moderate | -61.68 | 0.4410 |
| methods | 5 | Founder Sustained | Late Joiner Sustained | -23.76 | 0.9470 |
| methods | 5 | Late Joiner Moderate | Late Joiner Sustained | 37.92 | 0.1931 |
| methods | 10 | Founder Moderate | Founder Sustained | -17.78 | 0.9930 |
| methods | 10 | Founder Moderate | Late Joiner Moderate | -126.22 | 0.0303 |
| methods | 10 | Founder Moderate | Late Joiner Sustained | -90.73 | 0.2408 |
| methods | 10 | Founder Sustained | Late Joiner Moderate | -108.44 | 0.1257 |
| methods | 10 | Founder Sustained | Late Joiner Sustained | -72.95 | 0.4963 |
| methods | 10 | Late Joiner Moderate | Late Joiner Sustained | 35.49 | 0.4058 |
| methods | 20 | Founder Sustained | Late Joiner Sustained | -106.14 | 0.1554 |

**Post-hoc Analysis: Tukey HSD**

Following the Two-Way ANOVA, a Tukey Honestly Significant Difference (HSD) test is used to conduct pairwise comparisons. This test determines which specific group means differ significantly while maintaining a constant family-wise error rate.

The studentized range statistic ($q$) for comparing two group means is defined as:

$$q = \frac{\bar{Y}_{max} - \bar{Y}_{min}}{\sqrt{\frac{MS_{residual}}{n}}} \tag{5}$$

Where:

- $\bar{Y}_{max}$ and $\bar{Y}_{min}$ are the larger and smaller means of the two groups being compared.
- $MS_{residual}$ is the Mean Square Error derived from the ANOVA residual.
- $n$ is the number of observations in each group (adjusted for unequal sample sizes via the harmonic mean).

A pair of groups is considered significantly different if the absolute difference between their means exceeds the Honestly Significant Difference (HSD):

$$HSD = q_{\alpha,k,df}\sqrt{\frac{MS_{residual}}{n}} \tag{6}$$

Where $q_{\alpha,k,df}$ is the critical value from the studentized range distribution based on the significance level ($\alpha = 0.05$), the number of groups ($k$), and the degrees of freedom ($df$) from the ANOVA residual.

# 2 Repository: 14

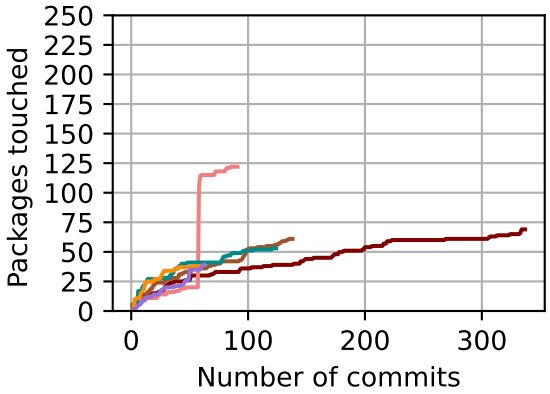
## 2.1 Time series sustained components touched by developer - 14 For developer commits and components touched.

Table 9: Table of first 6 sustained founder developers in the Scatter Developer graphs.

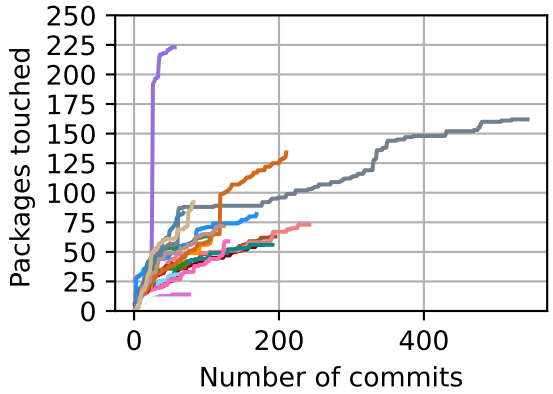
| Developer ID | Colour |
|---|---|
| 2930 | Maroon |
| 2932 | Sienna |
| 2934 | LightCoral |
| 2935 | DarkCyan |
| 2946 | DarkOrange |
| 2959 | MediumPurple |

Table 10: Table of first 21 sustained later joiner developers in the Scatter Developer graphs.

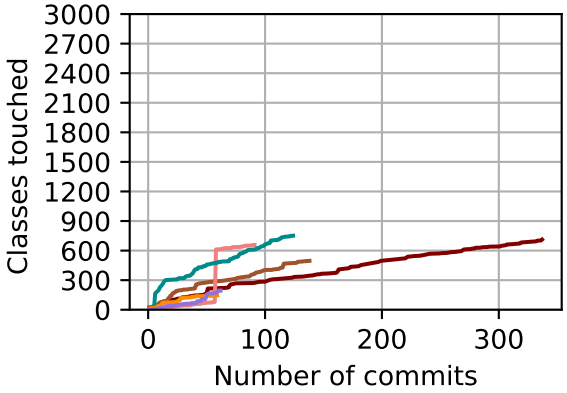
| **Developer ID** | **Colour** |
|---:|---|
| 1774 | Maroon |
| 2986 | Sienna |
| 3000 | LightCoral |
| 3020 | DarkCyan |
| 3037 | DarkOrange |
| 3050 | MediumPurple |
| 3071 | LightSkyBlue |
| 3072 | HotPink |
| 3141 | ForestGreen |
| 3143 | FireBrick |
| 3163 | DodgerBlue |
| 3172 | Tomato |
| 3178 | SeaGreen |
| 3216 | SlateGray |
| 3217 | Ivory |
| 3272 | Orchid |
| 3339 | Chocolate |
| 3354 | SteelBlue |
| 3456 | RosyBrown |
| 3473 | Azure |
| 3485 | Tan |



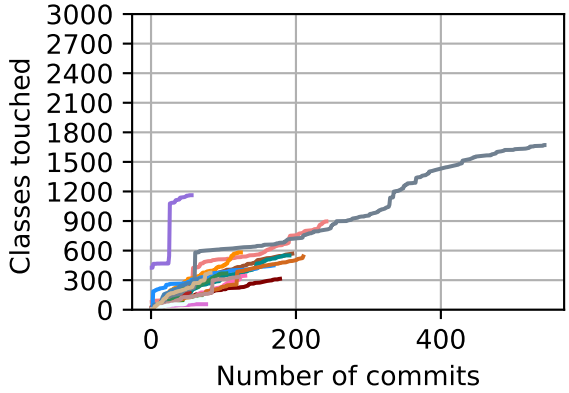

(a) Packages for sustained founder



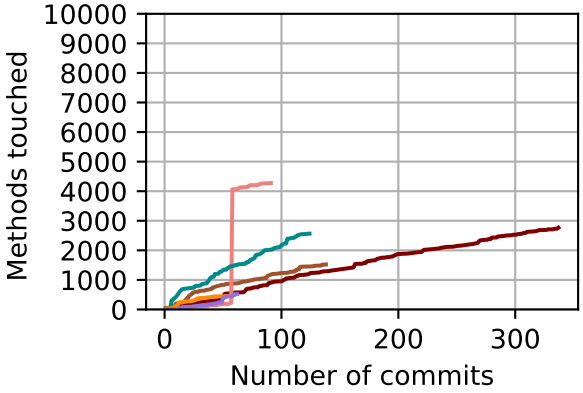

(b) Packages for sustained later joiner



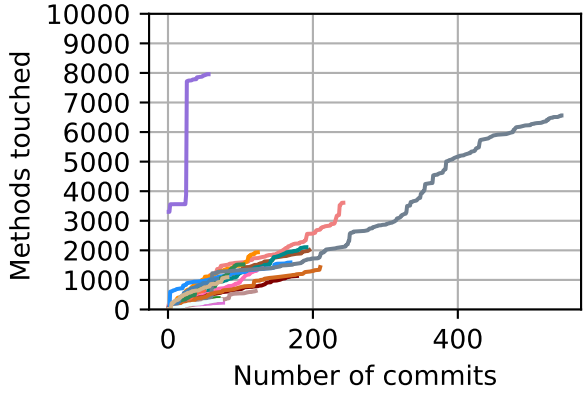

(c) Classes for sustained founder




(d) Classes for sustained later joiner



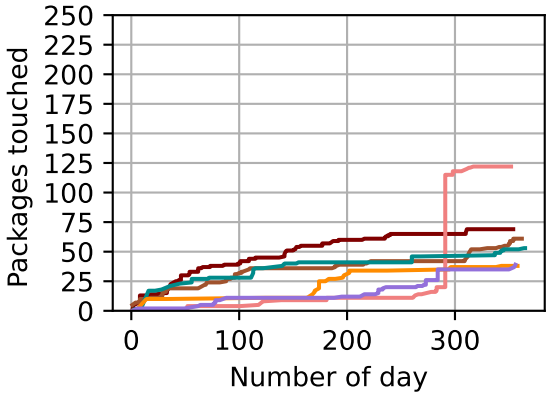

(e) Methods for sustained founder



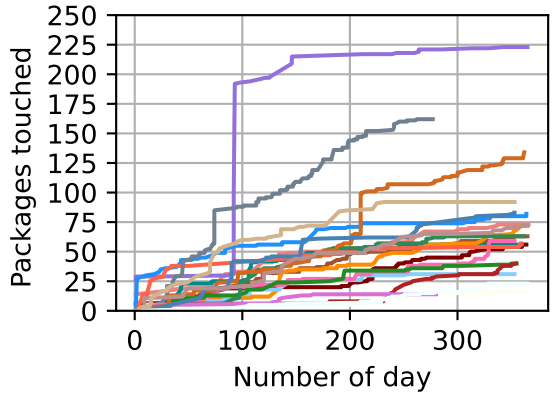

(f) Methods for sustained later joiner

Figure 40: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 9. Sustained later joiner developers with colours are shown in Table 10.

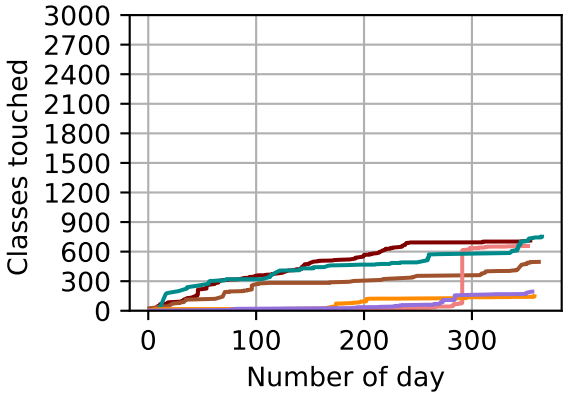
## 2.2 Time series sustained components touched by developer - 14 For developer day and components touched.



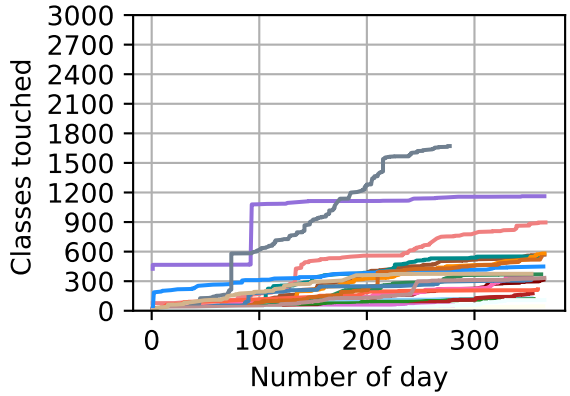

(a) Packages for sustained founder



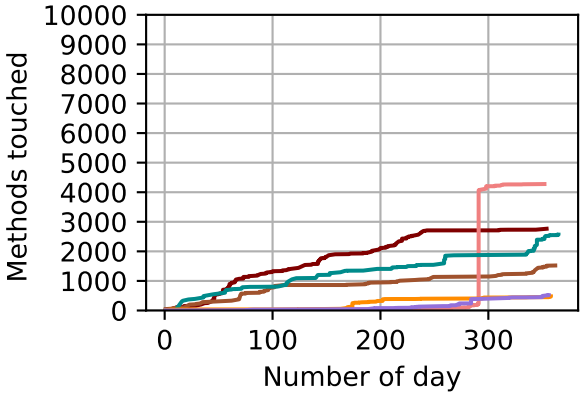

(b) Packages for sustained later joiner



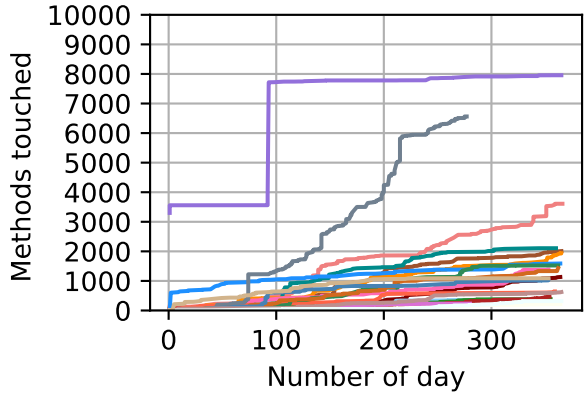

(c) Classes for sustained founder




(d) Classes for sustained later joiner




(e) Methods for sustained founder




(f) Methods for sustained later joiner

Figure 41: Repo: 14. First six (6) sustained founder and first twenty-one (21) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 9. Sustained later joiner developers with colours are shown in Table 10.

# 3 Repository: 21

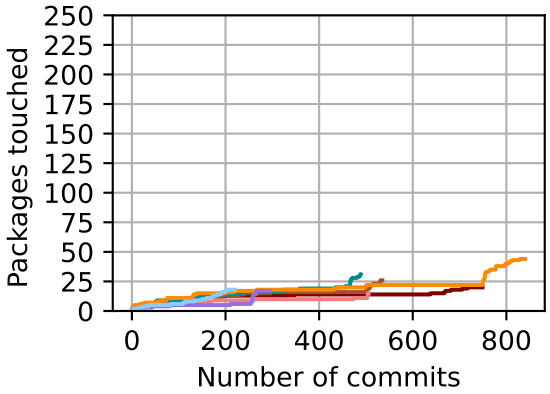
## 3.1 Time series sustained components touched by developer - 21 For developer commits and components touched.

Table 11: Table of first 7 sustained founder developers in the Scatter Developer graphs.

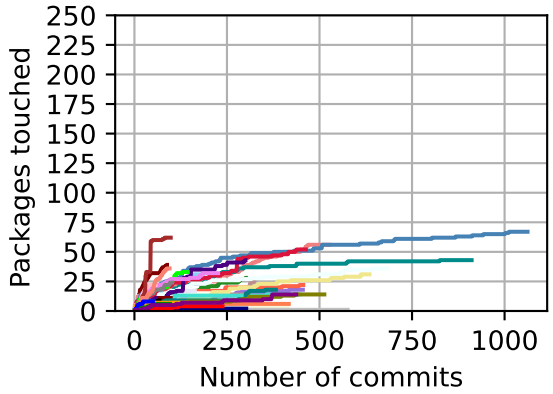
| Developer ID | Colour |
|---|---|
| 6245 | Maroon |
| 6246 | Sienna |
| 6247 | LightCoral |
| 6249 | DarkCyan |
| 6250 | DarkOrange |
| 6251 | MediumPurple |
| 6257 | LightSkyBlue |

Table 12: Table of first 50 sustained later joiner developers in the Scatter Developer graphs.

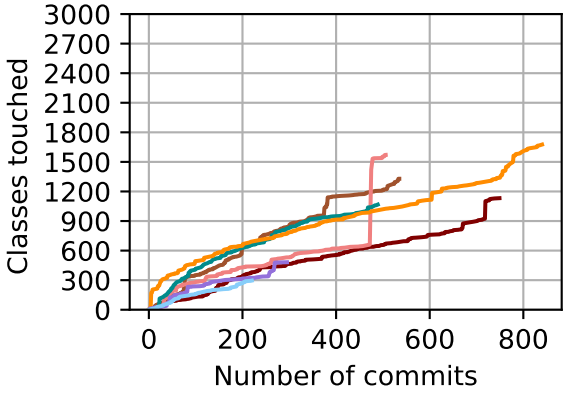
| Developer ID | Colour |
|---|---|
| 199 | Maroon |
| 6252 | Sienna |
| 6253 | LightCoral |
| 6259 | DarkCyan |
| 6260 | DarkOrange |
| 6262 | MediumPurple |
| 6263 | LightSkyBlue |
| 6264 | HotPink |
| 6267 | ForestGreen |
| 6271 | FireBrick |
| 6272 | DodgerBlue |
| 6276 | Tomato |
| 6278 | SeaGreen |
| 6280 | SlateGray |
| 6282 | Ivory |
| 6284 | Orchid |
| 6288 | Chocolate |
| 6295 | SteelBlue |
| 6297 | RosyBrown |
| 6298 | Azure |
| 6302 | Tan |
| 6306 | Plum |
| 6309 | Crimson |
| 6312 | Khaki |
| 6313 | Lavender |
| 6314 | Indigo |
| 6315 | Turquoise |
| 6326 | Beige |
| 6329 | Silver |
| 6333 | Gold |
| 6338 | Coral |
| 6342 | Lime |
| 6343 | Maroon |
| 6349 | Navy |
| 6352 | Teal |
| 6355 | Olive |
| 6363 | Violet |
| 6368 | Gray |
| 6369 | Pink |
| 6370 | Brown |
| 6373 | Black |
| 6374 | Yellow |
| 6375 | Magenta |
| 6379 | Cyan |
| 6391 | Salmon |
| 6410 | Orange |
| 6411 | Red |
| 6417 | Green |
| 6418 | Blue |
| 6420 | Purple |



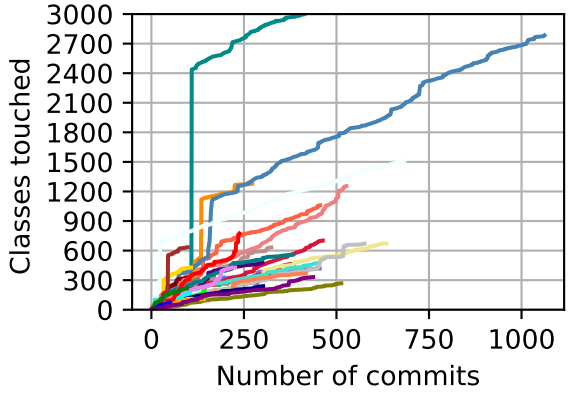

(a) Packages for sustained founder



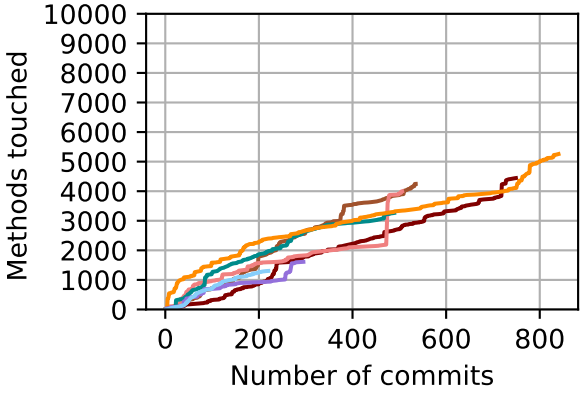

(b) Packages for sustained later joiner



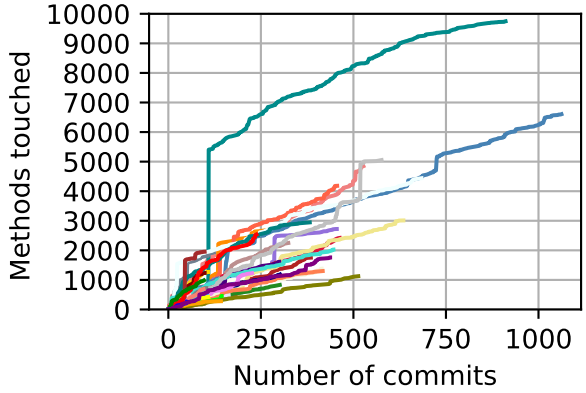

(c) Classes for sustained founder




(d) Classes for sustained later joiner



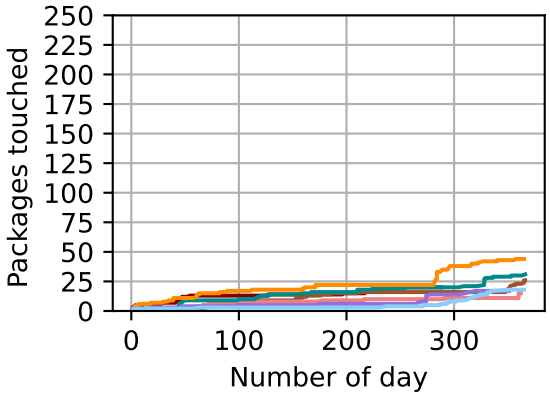

(e) Methods for sustained founder



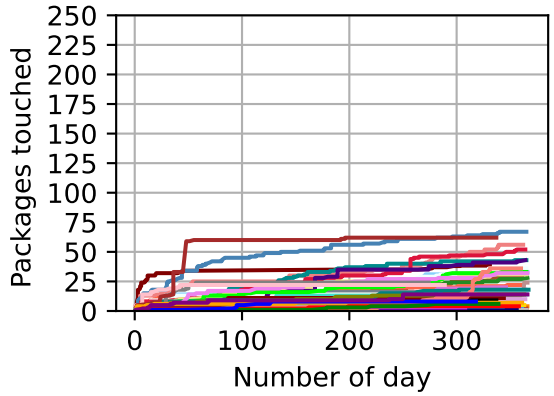

(f) Methods for sustained later joiner

Figure 42: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of commits (x-axis). Sustained founder developers with colours are shown in Table 11. Sustained later joiner developers with colours are shown in Table 12.

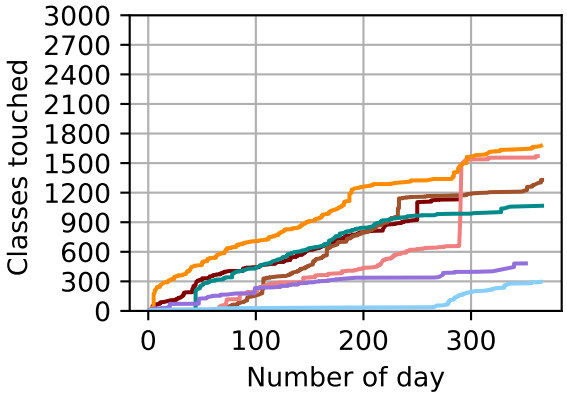
### 3.2 Time series sustained components touched by developer - 21 For developer day and components touched.



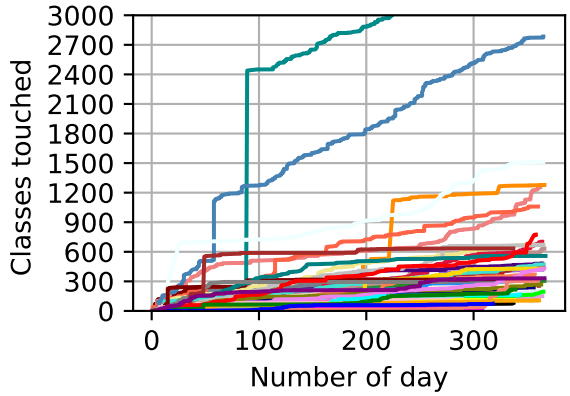

(a) Packages for sustained founder



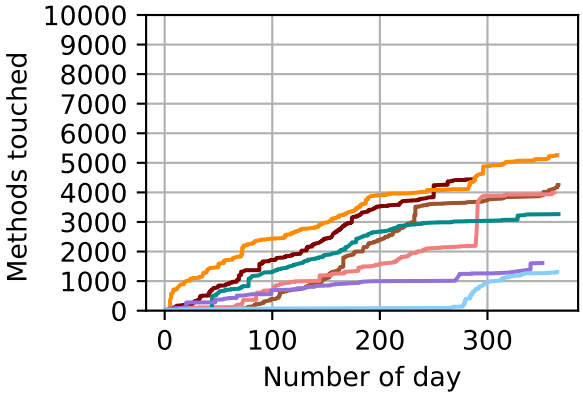

(b) Packages for sustained later joiner



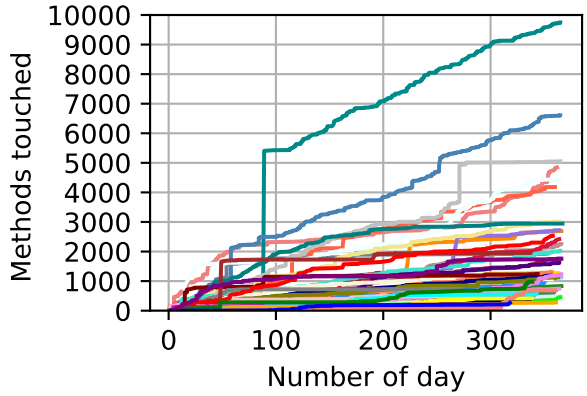

(c) Classes for sustained founder




(d) Classes for sustained later joiner




(e) Methods for sustained founder




(f) Methods for sustained later joiner

Figure 43: Repo: 21. First seven (7) sustained founder and first fifty (50) sustained later joiner developers. The number of components touched (y-axis) against the number of day (x-axis). Sustained founder developers with colours are shown in Table 11. Sustained later joiner developers with colours are shown in Table 12.